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Chapter 1

DC Circuit Analysis

1.1 Circuit Analysis Methods

1.1.1 Nodal method

1. Motivation : Nodal analysis provides a general procedure for analyzing circuits using **node voltages** as the circuit variables. Choosing node voltages **instead of element voltages** as circuit variables is convenient and reduces the number of equations one must solve simultaneously.
2. Steps to determine node voltages
 - (a) **Step 1. Node labeling.** Select a node as the **reference node**. Assign voltages $v_1, v_2, \dots, v_{n-1}$ to the remaining $n - 1$ nodes, which we might call **nonreference nodes**. These node voltages are referenced with respect to the **reference node** having zero potential. Determining the values of nonreference voltages is the goal of nodal analysis.
 - (b) **Step 2. Setting up a system of linear equations using KCL** Apply **KCL** to each of the $n - 1$ **nonreference nodes**. Use Ohm's law to express the branch currents in terms of node voltages. The number of **nonreference nodes** is equal to the number of independent equations that we will derive.
 - (c) **Step 3. Solving the system of equations.** Solve the resulting simultaneous equations to obtain the unknown node voltages.
3. Three types of reference nodes(ground)

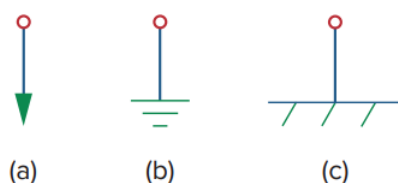
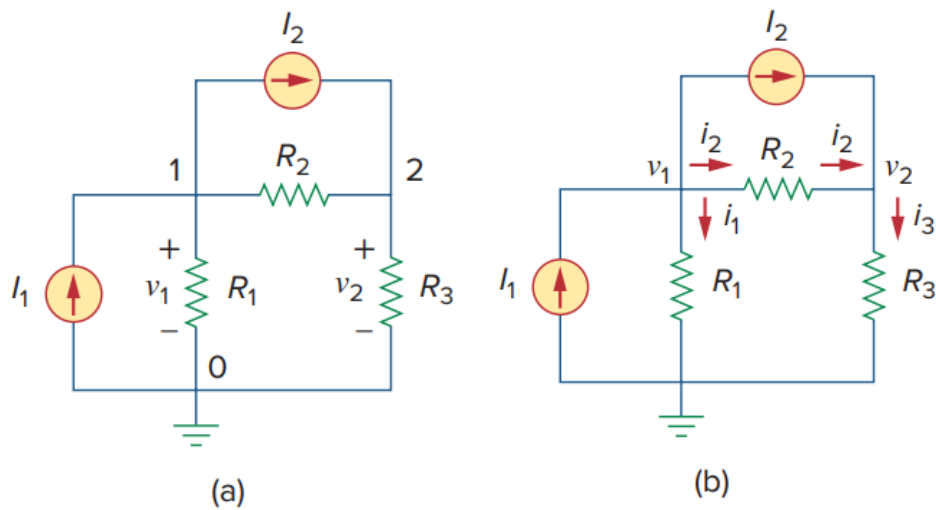


Figure 3.1

Common symbols for indicating a reference node, (a) common ground, (b) ground, (c) chassis ground.

Case 1. Simplest example without voltage sources (no supernodes)

- Step 1.** In the figure (a) above, step 1 is applied. Node 0 is the reference node ($v = 0$), and Node 1 and 2 are nonreference voltages v_1, v_2 . Each node voltage is the voltage rise from the reference node to the corresponding nonreference node or simply the voltage of that node with respect to the reference node.
- Step 2.** In the figure (b), we apply KCL at each nonreference node.
At node 1,

$$I_1 = i_1 + i_2$$

At node 2,

$$I_2 = i_2 + i_3$$

We now apply Ohm's law to express the unknown currents i_1, i_2 , and i_3 in terms of node voltages v_1, v_2 . The key idea to bear in mind is that, since resistance is a **passive** element, by the passive sign convention, current must always flow from a higher potential to a lower potential.

$$i = \frac{v_{\text{higher}} - v_{\text{lower}}}{R}$$

As a result,

$$\begin{aligned} i_1 &= \frac{v_1 - 0}{R_1} & \text{or} & \quad i_1 = G_1 v_1 \\ i_2 &= \frac{v_1 - v_2}{R_2} & \text{or} & \quad i_2 = G_2 (v_1 - v_2) \\ i_3 &= \frac{v_2 - 0}{R_3} & \text{or} & \quad i_3 = G_3 v_2 \end{aligned}$$

- Step 3.** Eliminating unknown currents i_1, i_2, i_3 in the KCL results

$$\begin{aligned} I_1 &= I_2 + \frac{v_1}{R_1} + \frac{v_1 - v_2}{R_2} \\ I_2 + \frac{v_1 - v_2}{R_2} &= \frac{v_2}{R_3} \end{aligned}$$

In terms of the conductances, these equations become

$$I_1 = I_2 + G_1 v_1 + G_2(v_1 - v_2)$$

$$I_2 + G_2(v_1 - v_2) = G_3 v_2$$

Our problem at hand is to determine unknown node voltages v_1, v_2 in terms of given conductances and current sources I_1, I_2 and thus is conveniently formulated in the matrix equations

$$\begin{bmatrix} G_1 + G_2 & -G_2 \\ -G_2 & G_2 + G_3 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} I_1 - I_2 \\ I_2 \end{bmatrix}$$

which can be solved using Kramer's method, inversion, or numerical linear algebra using computer.

Remark. Observe that each of the diagonal terms is the sum of the conductances connected directly to node 1 or 2, while the off-diagonal terms are the negatives of the conductances connected between the nodes. Also, each term on the right-hand side of the above matrix equation is the algebraic sum of the currents entering the node.

4. **Numerical example.** Determine the voltages at the nodes in Fig. 3.5(a)

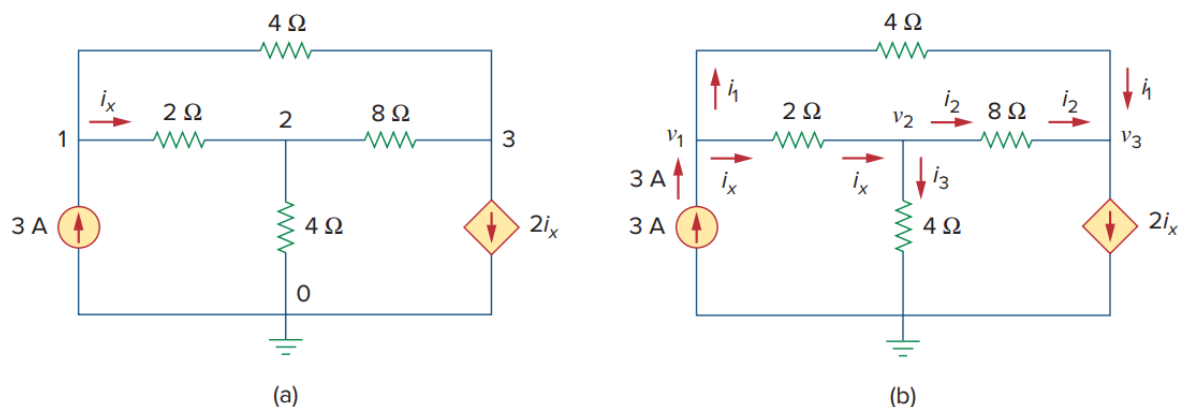


Figure 3.5

For Example 3.2: (a) original circuit, (b) circuit for analysis.

Solution: The circuit in this example has three nonreference nodes, unlike the previous example which has two nonreference nodes. We assign voltages to the three nodes as shown in Fig. 3.5(b) and label the currents.

At node 1,

$$3 = i_1 + i_x \Rightarrow 3 = \frac{v_1 - v_3}{4} + \frac{v_1 - v_2}{2}$$

Multiplying by 4 and rearranging terms, we get

$$3v_1 - 2v_2 - v_3 = 12 \quad (3.2.1)$$

At node 2,

$$i_x = i_2 + i_3 \Rightarrow \frac{v_1 - v_2}{2} = \frac{v_2 - v_3}{8} + \frac{v_2 - 0}{4}$$

Multiplying by 8 and rearranging terms, we get

$$-4v_1 + 7v_2 - v_3 = 0 \quad (3.2.2)$$

At node 3,

$$i_1 + i_2 = 2i_x \Rightarrow \frac{v_1 - v_3}{4} + \frac{v_2 - v_3}{8} = \frac{2(v_1 - v_2)}{2}$$

Multiplying by 8, rearranging terms, and dividing by 3, we get

$$2v_1 - 3v_2 + v_3 = 0 \quad (3.2.3)$$

We have three simultaneous equations to solve to get the node voltages v_1 , v_2 , and v_3 . We can solve the equations in three ways (elimination, Cramer's rule, MATLAB-based matrix inversion). Here, we only introduce solution procedure using Cramer's rule : To use Cramer's rule, we put the above system of linear equations in matrix form.

$$\begin{bmatrix} 3 & -2 & -1 \\ -4 & 7 & -1 \\ 2 & -3 & 1 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix} = \begin{bmatrix} 12 \\ 0 \\ 0 \end{bmatrix}$$

From this, we obtain

$$v_1 = \frac{\Delta_1}{\Delta}, \quad v_2 = \frac{\Delta_2}{\Delta}, \quad v_3 = \frac{\Delta_3}{\Delta}$$

where Δ , Δ_1 , Δ_2 , and Δ_3 are the determinants to be calculated as follows.

$$\Delta = \begin{vmatrix} 3 & -2 & -1 \\ -4 & 7 & -1 \\ 2 & -3 & 1 \end{vmatrix} = \begin{vmatrix} 3 & -2 & -1 \\ -4 & 7 & -1 \\ 2 & -3 & 1 \end{vmatrix} = \begin{vmatrix} 3 & -2 & -1 \\ -4 & 7 & -1 \\ 2 & -3 & 1 \end{vmatrix} = 21 - 12 + 4 + 14 - 9 - 8 = 10$$

Similarly, we obtain

$$\Delta_1 = \begin{vmatrix} 12 & -2 & -1 \\ 0 & 7 & -1 \\ 0 & -3 & 1 \end{vmatrix} = 84 + 0 + 0 - 0 - 36 - 0 = 48$$

$$\Delta_2 = \begin{vmatrix} 3 & 12 & -1 \\ -4 & 0 & -1 \\ 2 & 0 & 1 \end{vmatrix} = 0 + 0 - 24 - 0 - 0 + 48 = 24$$

$$\Delta_3 = \begin{vmatrix} 3 & -2 & 12 \\ -4 & 7 & 0 \\ 2 & -3 & 0 \end{vmatrix} = 0 + 144 + 0 - 168 - 0 - 0 = -24$$

$$\therefore v_1 = \frac{\Delta_1}{\Delta} = \frac{48}{10} = 4.8 \text{ V}, \quad v_2 = \frac{\Delta_2}{\Delta} = \frac{24}{10} = 2.4 \text{ V}, \quad v_3 = \frac{\Delta_3}{\Delta} = \frac{-24}{10} = -2.4 \text{ V}$$

Case 2. Nodal analysis with Voltage sources : Supernode treatment

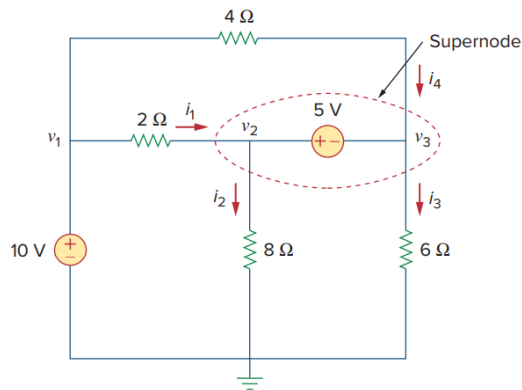


Figure 3.7
A circuit with a supernode.

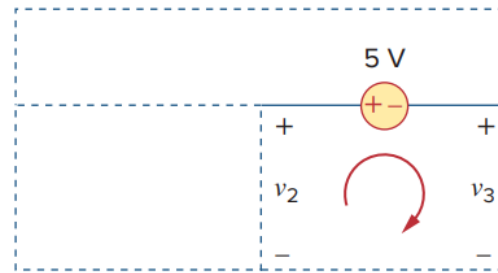


Figure 3.8
Applying KVL to a supernode.

Voltage sources in the nodal analysis requires a special care as follows :

1. If a voltage source is connected between the reference node and a nonreference node, we simply set the voltage at the nonreference node equal to the voltage of the voltage source. In Fig. 3.7, for example,

$$v_1 = 10V \quad (1)$$

Thus, our analysis is somewhat simplified by this knowledge of the voltage at this node.

2. If the voltage source (dependent or independent) is connected between two nonreference nodes, the two nonreference nodes form a generalized node or supernode; we apply both KCL and KVL to determine the node voltages. Note that

A **supernode** is formed by enclosing a (dependent or independent) voltage source connected between two nonreference nodes and any elements connected in parallel with it.

A supernode may be regarded as a closed surface enclosing the voltage source and its two nodes.

In Fig. 3.7, nodes 2 and 3 form a supernode.

3. We analyze a circuit with supernodes using the same three steps mentioned in the previous section **except** that the supernodes are treated differently.
(**Important.** Why? Because an essential component of nodal analysis is applying KCL, which requires knowing the current through each element. There is no way of knowing the current through a voltage source in advance.)
4. However, KCL must be satisfied at a supernode like any other node. Hence, at the supernode,

$$i_1 + i_4 = i_2 + i_3$$

$$\frac{v_1 - v_2}{2} + \frac{v_1 - v_3}{4} = \frac{v_2 - 0}{8} + \frac{v_3 - 0}{6} \quad (2)$$

To apply Kirchhoff's voltage law to the supernode in Fig. 3.7, we redraw the circuit as shown in Fig. 3.8. Going around the loop in the clockwise direction gives

$$-v_2 + 5 + v_3 = 0 \Rightarrow v_2 - v_3 = 5 \quad (3)$$

We obtain node voltages from the three equations (1), (2), (3).

5. Remember the following properties of the supernode :

1. The voltage source inside the supernode provides a constraint equation needed to solve for the node voltages.
2. A supernode has no voltage of its own.
3. A supernode requires the application of both KCL and KVL.

6. Example with dependent voltage sources

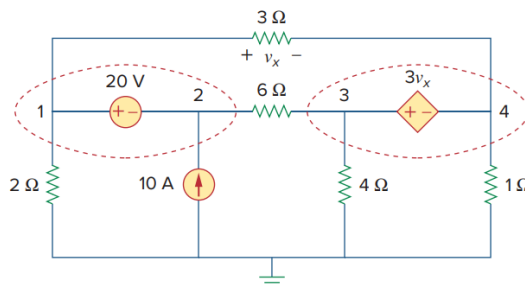


Figure 3.12
For Example 3.4.

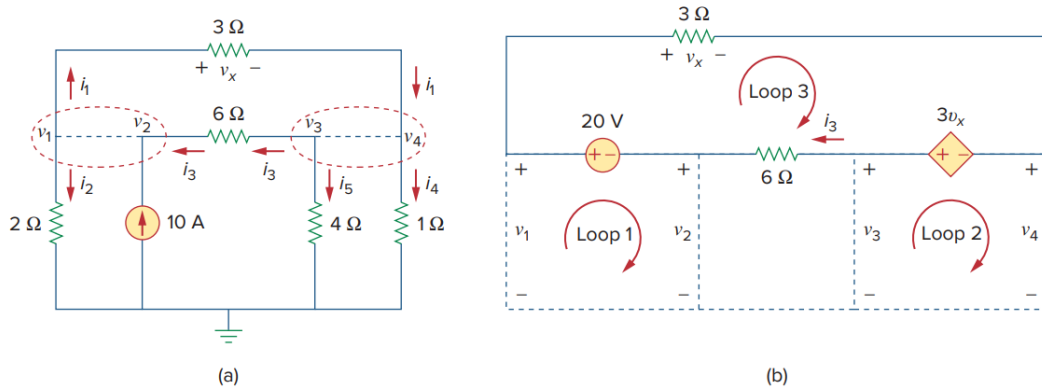


Figure 3.13
Applying: (a) KCL to the two supernodes, (b) KVL to the loops.

We apply KCL to the two supernodes:

$$\begin{aligned} \text{At supernode 1-2: } i_3 + 10 &= i_1 + i_2 \\ \Rightarrow \frac{v_3 - v_2}{6} + 10 &= \frac{v_1 - v_4}{3} + \frac{v_1}{2} \\ \Rightarrow 5v_1 + v_2 - v_3 - 2v_4 &= 60 \end{aligned} \quad (3.4.1)$$

$$\begin{aligned}
\text{At supernode 3-4: } i_1 &= i_3 + i_4 + i_5 \\
\Rightarrow \frac{v_1 - v_4}{3} &= \frac{v_3 - v_2}{6} + v_4 + \frac{v_3}{4} \\
\Rightarrow 4v_1 + 2v_2 - 5v_3 - 16v_4 &= 0
\end{aligned} \tag{3.4.2}$$

Apply KVL to the loops:

$$\text{Loop 1: } -v_1 + 20 + v_2 = 0 \Rightarrow v_1 - v_2 = 20 \tag{3.4.3}$$

$$\begin{aligned}
\text{Loop 2: } -v_3 + 3v_x + v_4 &= 0, \quad v_x = v_1 - v_4 \\
\Rightarrow 3v_1 - v_3 - 2v_4 &= 0
\end{aligned} \tag{3.4.4}$$

$$\begin{aligned}
\text{Loop 3: } v_x - 3v_x + 6i_3 - 20 &= 0 \\
\text{with } 6i_3 &= v_3 - v_2, \quad v_x = v_1 - v_4 \\
\Rightarrow -2v_1 - v_2 + v_3 + 2v_4 &= 20
\end{aligned} \tag{3.4.5}$$

Using $v_2 = v_1 - 20$ from (3.4.3), substitute into (3.4.1) and (3.4.2):

$$6v_1 - v_3 - 2v_4 = 80 \tag{3.4.6}$$

$$6v_1 - 5v_3 - 16v_4 = 40 \tag{3.4.7}$$

Combine with (3.4.4):

$$\begin{bmatrix} 3 & -1 & -2 \\ 6 & -1 & -2 \\ 6 & -5 & -16 \end{bmatrix} \begin{bmatrix} v_1 \\ v_3 \\ v_4 \end{bmatrix} = \begin{bmatrix} 0 \\ 80 \\ 40 \end{bmatrix}$$

Using Cramer's rule:

$$\Delta = \begin{vmatrix} 3 & -1 & -2 \\ 6 & -1 & -2 \\ 6 & -5 & -16 \end{vmatrix} = -18$$

$$\Delta_1 = \begin{vmatrix} 0 & -1 & -2 \\ 80 & -1 & -2 \\ 40 & -5 & -16 \end{vmatrix} = -480 \Rightarrow v_1 = \frac{-480}{-18} = 26.67 \text{ V}$$

$$\Delta_3 = \begin{vmatrix} 3 & 0 & -2 \\ 6 & 80 & -2 \\ 6 & 40 & -16 \end{vmatrix} = -3120 \Rightarrow v_3 = \frac{-3120}{-18} = 173.33 \text{ V}$$

$$\Delta_4 = \begin{vmatrix} 3 & -1 & 0 \\ 6 & -1 & 80 \\ 6 & -5 & 40 \end{vmatrix} = 840 \Rightarrow v_4 = \frac{840}{-18} = -46.67 \text{ V}$$

$$v_2 = v_1 - 20 = 6.667 \text{ V}$$

1.1.2 Mesh method

1. A **mesh** is a loop that does not contain any other loops within it

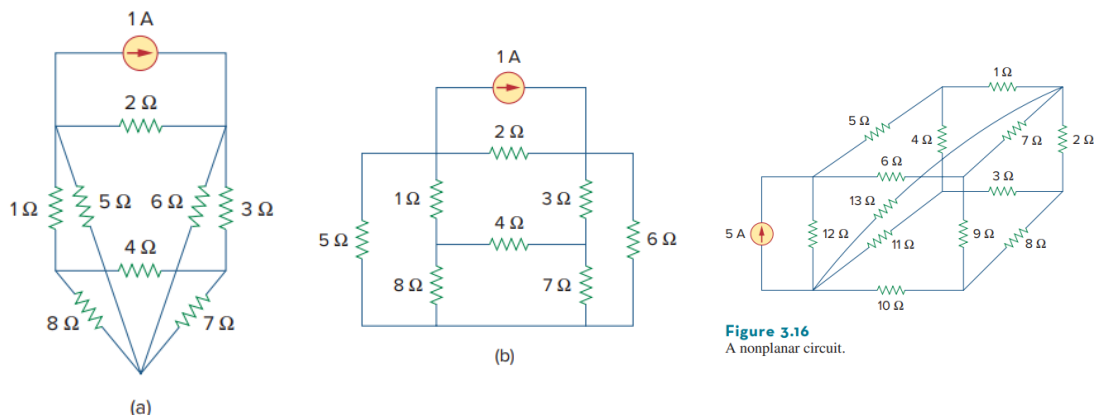


Figure 3.16
A nonplanar circuit.

2. **Mesh analysis** can be applied only to **planar** circuits. (See Figures above) The left circuit with crossing branches can be reduced to a planar circuit in the middle and thus is amenable to nodal analysis. However, Fig. 3.16. is nonplanar.
3. Three steps to apply mesh analysis

Steps to Determine Mesh Currents

- (a) Assign mesh currents $i_1, i_2, \dots, i_n$ to the n meshes.
- (b) Apply KVL to each of the n meshes. Use Ohm's law to express the voltages in terms of the mesh currents.
- (c) Solve the resulting n simultaneous equations to get the mesh currents.

4. **Case 1. Simplest example without current sources**

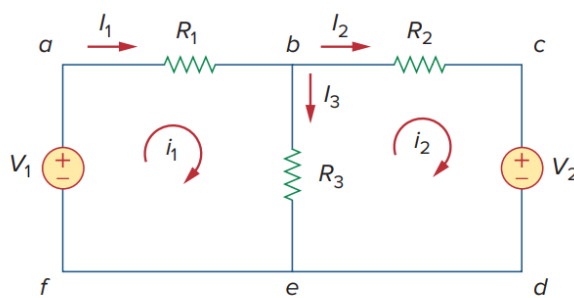


Figure 3.17
A circuit with two meshes.

Step 1: Assign mesh currents

Assign mesh currents i_1 and i_2 to the two meshes. Although any direction can be chosen, we conventionally assume clockwise mesh currents.

Step 2: Apply KVL to each mesh

For Mesh 1:

$$\begin{aligned} -V_1 + R_1 i_1 + R_3(i_1 - i_2) &= 0 \\ (R_1 + R_3)i_1 - R_3 i_2 &= V_1 \end{aligned} \quad (3.13)$$

For Mesh 2:

$$\begin{aligned} R_2 i_2 + V_2 + R_3(i_2 - i_1) &= 0 \\ -R_3 i_1 + (R_2 + R_3)i_2 &= -V_2 \end{aligned} \quad (3.14)$$

Step 3: Solve the resulting system

Writing Eqs. (3.13) and (3.14) in matrix form:

$$\begin{bmatrix} R_1 + R_3 & -R_3 \\ -R_3 & R_2 + R_3 \end{bmatrix} \begin{bmatrix} i_1 \\ i_2 \end{bmatrix} = \begin{bmatrix} V_1 \\ -V_2 \end{bmatrix} \quad (3.15)$$

Solving the above linear system gives the mesh currents i_1 and i_2 .

5. Remarks on mesh analysis method

- (a) **Quick setup of matrix equations** : We notice that each of the diagonal terms is the sum of the resistances in the related mesh, while each of the off-diagonal terms is the negative of the resistance common to meshes 1 and 2. Each term on the right-hand side of Eq.(3.15.) is the algebraic sum taken clockwise of all independent voltage sources in the related mesh.
- (b) **Number of Mesh Equations**: According to the fundamental theorem of network topology, if a circuit has n nodes, b branches, and l independent loops or meshes, then

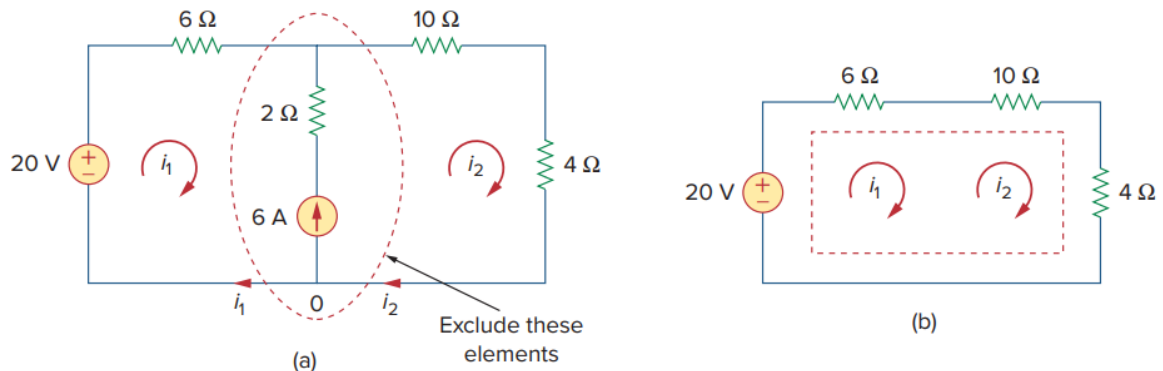
$$l = b - n + 1.$$

Hence, l independent simultaneous equations are required to solve the circuit using mesh analysis.

- (c) **Branch Currents vs. Mesh Currents**: Branch currents are different from mesh currents unless the mesh is isolated. We use i for a mesh current and I for a branch current. The current elements I_1 , I_2 , and I_3 are algebraic sums of the mesh currents:

$$I_1 = i_1, \quad I_2 = i_2, \quad I_3 = i_1 - i_2 \quad (3.16)$$

6. Case 2. Mesh analysis with current sources

**Figure 3.23**

(a) Two meshes having a current source in common, (b) a supermesh, created by excluding the current source.

- (a) **Supermesh.** When a current source exists between two meshes: Consider the circuit in Fig. 3.23(a), for example. We create a **supermesh** by excluding the current source and any elements connected in series with it, as shown in Fig. 3.23(b). Thus,

A supermesh results when two meshes have a (dependent or independent) current source in common.

As shown in Fig. 3.23(b), we create a supermesh as the periphery of the two meshes and treat it differently. (If a circuit has two or more supermeshes that intersect, they should be combined to form a larger supermesh.)

Why treat the supermesh differently? Because mesh analysis applies KVL—which requires that we know the voltage across each branch—and we do not know the voltage across a current source in advance.

- (b) However, a supermesh must satisfy KVL like any other mesh. Therefore, applying KVL to the supermesh in Fig. 3.23(b) gives

$$-20 + 6i_1 + 10i_2 + 4i_2 = 0$$

We apply KCL to a node in the branch where the two meshes intersect. Applying KCL to node 0 in Fig. 3.23(a) gives

$$i_2 = i_1 + 6 \quad (3.19)$$

Solving Eqs. (3.18) and (3.19), we get

$$i_1 = -3.2 \text{ A}, \quad i_2 = 2.8 \text{ A} \quad (3.20)$$

- (c) Note the following properties of a supermesh:
- i. The current source in the supermesh provides the constraint equation necessary to solve for the mesh currents.
 - ii. A supermesh has no current of its own.
 - iii. A supermesh requires the application of both KVL and KCL.

7. Case 3. Supermesh Analysis with Dependent and Independent Sources

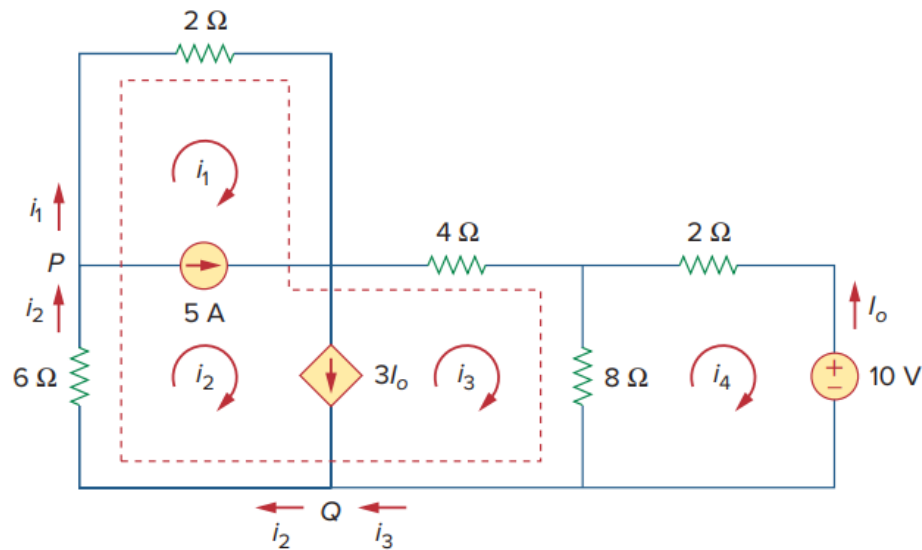


Figure 3.24
For Example 3.7.

Note that meshes 1 and 2 form a supermesh because they have an independent current source in common. Also, meshes 2 and 3 form another supermesh because they have a dependent current source in common. The two supermeshes intersect and form a larger supermesh as shown.

Applying KVL to the larger supermesh:

$$\begin{aligned} 2i_1 + 4i_3 + 8(i_3 - i_4) + 6i_2 &= 0 \\ i_1 + 3i_2 + 6i_3 - 4i_4 &= 0 \end{aligned} \quad (3.7.1)$$

For the independent current source, applying KCL to node P:

$$i_2 = i_1 + 5 \quad (3.7.2)$$

For the dependent current source, applying KCL to node Q:

$$\begin{aligned} i_2 &= i_3 + 3I_o \\ \text{But } I_o &= -i_4 \Rightarrow i_2 = i_3 - 3i_4 \end{aligned} \quad (3.7.3)$$

Applying KVL in mesh 4:

$$\begin{aligned} 2i_4 + 8(i_4 - i_3) + 10 &= 0 \\ 5i_4 - 4i_3 &= -5 \end{aligned} \quad (3.7.4)$$

Solving Eqs. (3.7.1) to (3.7.4), we get:

$$i_1 = -7.5 \text{ A}, \quad i_2 = -2.5 \text{ A}, \quad i_3 = 3.93 \text{ A}, \quad i_4 = 2.143 \text{ A}$$

1.1.3 Quick inspection of circuits

Quick nodal analysis in the absence of dependent sources

$$\begin{bmatrix} G_{11} & G_{12} & \cdots & G_{1N} \\ G_{21} & G_{22} & \cdots & G_{2N} \\ \vdots & \vdots & \ddots & \vdots \\ G_{N1} & G_{N2} & \cdots & G_{NN} \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_N \end{bmatrix} = \begin{bmatrix} i_1 \\ i_2 \\ \vdots \\ i_N \end{bmatrix} \quad (3.22)$$

or simply,

$$\mathbf{G}\mathbf{v} = \mathbf{i} \quad (3.23)$$

where

G_{kk} = Sum of the conductances connected to node k

$G_{kj} = G_{jk}$ = Negative of the sum of the conductances directly connecting nodes k and j , $k \neq j$

v_k = Unknown voltage at node k

i_k = Sum of all independent current sources directly connected to node k

i_k is positive if entering this node.

$\mathbf{G}$ is called the **conductance matrix**; $\mathbf{v}$ is the output vector; and $\mathbf{i}$ is the input vector. Equation (3.22) can be solved to obtain the unknown node voltages.

Limit of Validity : This is valid for circuits with only independent current sources and linear resistors.

Example.

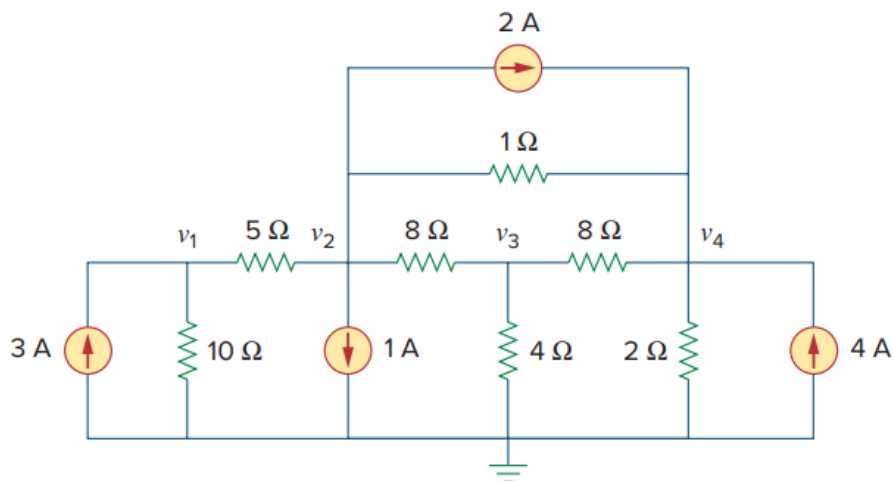


Figure 3.27
For Example 3.8.

Solution:

The circuit in Fig. 3.27 has four nonreference nodes, so we need four node equations. This implies that the size of the conductance matrix $\mathbf{G}$ is 4×4 . The diagonal terms of $\mathbf{G}$, in siemens, are

$$\begin{aligned} G_{11} &= \frac{1}{5} + \frac{1}{10} = 0.3, \\ G_{22} &= \frac{1}{5} + \frac{1}{8} + \frac{1}{1} = 1.325, \\ G_{33} &= \frac{1}{8} + \frac{1}{8} + \frac{1}{4} = 0.5, \\ G_{44} &= \frac{1}{8} + \frac{1}{2} + \frac{1}{1} = 1.625. \end{aligned}$$

The off-diagonal terms are

$$\begin{aligned} G_{12} &= G_{21} = -\frac{1}{5} = -0.2, \\ G_{13} &= G_{14} = G_{31} = G_{41} = 0, \\ G_{23} &= G_{32} = -\frac{1}{8} = -0.125, \\ G_{24} &= G_{42} = -\frac{1}{1} = -1, \\ G_{34} &= G_{43} = -\frac{1}{8} = -0.125. \end{aligned}$$

The input current vector $\mathbf{i}$ has the following terms, in amperes:

$$\begin{aligned} i_1 &= 3, \\ i_2 &= -1 - 2 = -3, \\ i_3 &= 0, \\ i_4 &= 2 + 4 = 6. \end{aligned}$$

Thus, the node-voltage equations are

$$\begin{bmatrix} 0.3 & -0.2 & 0 & 0 \\ -0.2 & 1.325 & -0.125 & -1 \\ 0 & -0.125 & 0.5 & -0.125 \\ 0 & -1 & -0.125 & 1.625 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \\ v_3 \\ v_4 \end{bmatrix} = \begin{bmatrix} 3 \\ -3 \\ 0 \\ 6 \end{bmatrix}$$

which can be solved using *MATLAB* to obtain the node voltages v_1, v_2, v_3 , and v_4 .

Quick mesh analysis without dependent sources

$$\begin{bmatrix} R_{11} & R_{12} & \cdots & R_{1N} \\ R_{21} & R_{22} & \cdots & R_{2N} \\ \vdots & \vdots & \ddots & \vdots \\ R_{N1} & R_{N2} & \cdots & R_{NN} \end{bmatrix} \begin{bmatrix} i_1 \\ i_2 \\ \vdots \\ i_N \end{bmatrix} = \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_N \end{bmatrix} \quad (3.25)$$

or simply,

$$\mathbf{R}\mathbf{i} = \mathbf{v} \quad (3.26)$$

where

R_{kk} = Sum of the resistances in mesh k

$R_{kj} = R_{jk}$ = Negative of the sum of the resistances in common with meshes k and j , $k \neq j$

i_k = Unknown mesh current in mesh k (clockwise direction)

v_k = Sum of all independent voltage sources in mesh k
(clockwise, with voltage rise treated as positive)

$\mathbf{R}$ is called the **resistance matrix**; $\mathbf{i}$ is the output vector; and $\mathbf{v}$ is the input vector. Equation (3.25) can be solved to obtain the unknown mesh currents.

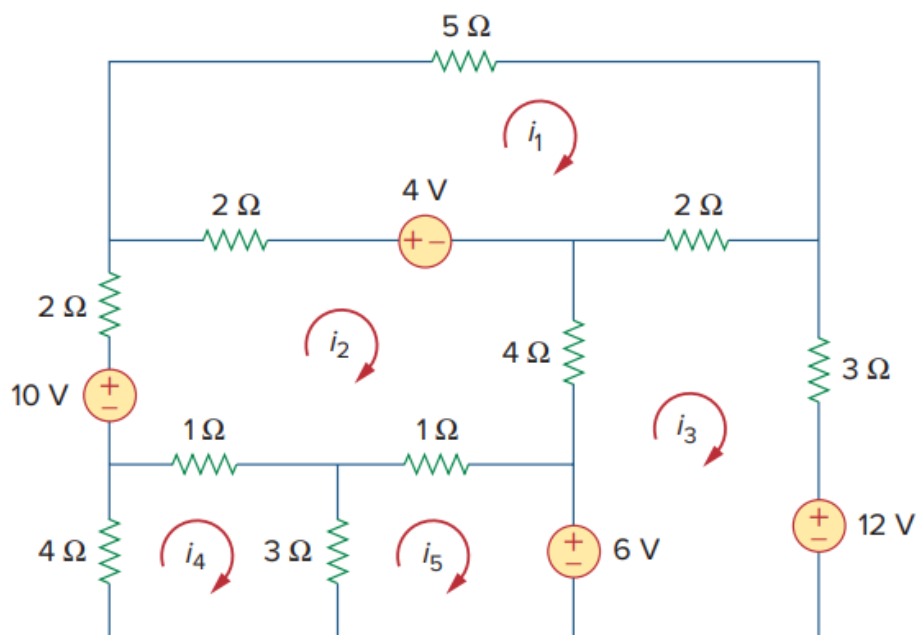
Example.

Figure 3.29
For Example 3.9.

Solution:

We have five meshes, so the resistance matrix is 5×5 . The diagonal terms, in ohms, are:

$$\begin{aligned} R_{11} &= 5 + 2 + 2 = 9, \\ R_{22} &= 2 + 4 + 1 + 1 + 2 = 10, \\ R_{33} &= 2 + 3 + 4 = 9, \\ R_{44} &= 1 + 3 + 4 = 8, \\ R_{55} &= 1 + 3 = 4. \end{aligned}$$

The off-diagonal terms are:

$$\begin{aligned} R_{12} &= R_{21} = -2, \\ R_{13} &= R_{31} = -2, \\ R_{14} &= R_{41} = 0, \\ R_{15} &= R_{51} = 0, \\ R_{23} &= R_{32} = -4, \\ R_{24} &= R_{42} = -1, \\ R_{25} &= R_{52} = -1, \\ R_{34} &= R_{43} = 0, \\ R_{35} &= R_{53} = 0, \\ R_{45} &= R_{54} = -3. \end{aligned}$$

The input voltage vector $\mathbf{v}$ has the following terms in volts:

$$\begin{aligned} v_1 &= 4, \\ v_2 &= 10 - 4 = 6, \\ v_3 &= -12 + 6 = -6, \\ v_4 &= 0, \\ v_5 &= -6. \end{aligned}$$

Thus, the mesh-current equations are:

$$\begin{bmatrix} 9 & -2 & -2 & 0 & 0 \\ -2 & 10 & -4 & -1 & -1 \\ -2 & -4 & 9 & 0 & 0 \\ 0 & -1 & 0 & 8 & -3 \\ 0 & -1 & 0 & -3 & 4 \end{bmatrix} \begin{bmatrix} i_1 \\ i_2 \\ i_3 \\ i_4 \\ i_5 \end{bmatrix} = \begin{bmatrix} 4 \\ 6 \\ -6 \\ 0 \\ -6 \end{bmatrix}$$

1.1.4 Comparison : Nodal vs Mesh method

Both nodal and mesh analyses provide a systematic way of analyzing a complex network. Someone may ask: Given a network to be analyzed, how do we know which method is better or more efficient? The choice of the better method is dictated by two factors.

- **The first factor is the nature of the particular network.** Networks that contain many series-connected elements, voltage sources, or supermeshes are more suitable for mesh analysis, whereas networks with parallel-connected elements, current sources, or supernodes are more suitable for nodal analysis.
Also, a circuit with fewer nodes than meshes is better analyzed using nodal analysis, while a circuit with fewer meshes than nodes is better analyzed using mesh analysis. The key is to select the method that results in the smaller number of equations.
- **The second factor is the information required.** If node voltages are required, it may be expedient to apply nodal analysis. If branch or mesh currents are required, it may be better to use mesh analysis.
- **It is helpful to be familiar with both methods of analysis, for at least two reasons.**
 - First, one method can be used to check the results from the other method, if possible.
 - Second, since each method has its limitations, only one method may be suitable for a particular problem. For example, mesh analysis is the only method to use in analyzing transistor circuits.
But mesh analysis cannot easily be used to solve an op amp circuit, because there is no direct way to obtain the voltage across the op amp itself.
- **For nonplanar networks, nodal analysis is the only option,** because mesh analysis only applies to planar networks.
- **Also, nodal analysis is more amenable to solution by computer,** as it is easy to program. This allows one to analyze complicated circuits that defy hand calculation. A computer software package based on nodal analysis is introduced next.

Here is the summary of the discussion above.

Condition	Prefer Nodal Analysis	Prefer Mesh Analysis
Topology of network	Parallel-connected elements Current sources Supernodes	Series-connected elements Voltage sources Supermeshes
Number of equations	Fewer nodes than meshes	Fewer meshes than nodes
Type of variable required	Node voltages	Branch or mesh currents
Planarity of circuit	Applicable to all networks (planar or non-planar)	Only applicable to planar networks
Ease of computer solution	Easier to program and solve computationally	Less suited for direct computer solution
Special application	Op-amp circuits (need voltage across terminals)	Transistor circuits (e.g., BJT mesh models)
Cross-verification	Can verify mesh analysis results	Can verify nodal analysis results

1.1.5 Superposition principle for linear circuits

Linear circuits

1. **Def. Linearity** is the property of an element describing a linear relationship between cause and effect. The property is a combination of both the homogeneity (scaling) property and the additivity property.
2. **Def. Linear circuit.** In general, a circuit is linear if it is both additive and homogeneous. In other words, a linear circuit is one whose output is linearly related (or directly proportional) to its input. A linear circuit consists of only linear elements, linear dependent sources, and independent sources.

Example. Using linearity to find the actual value of current

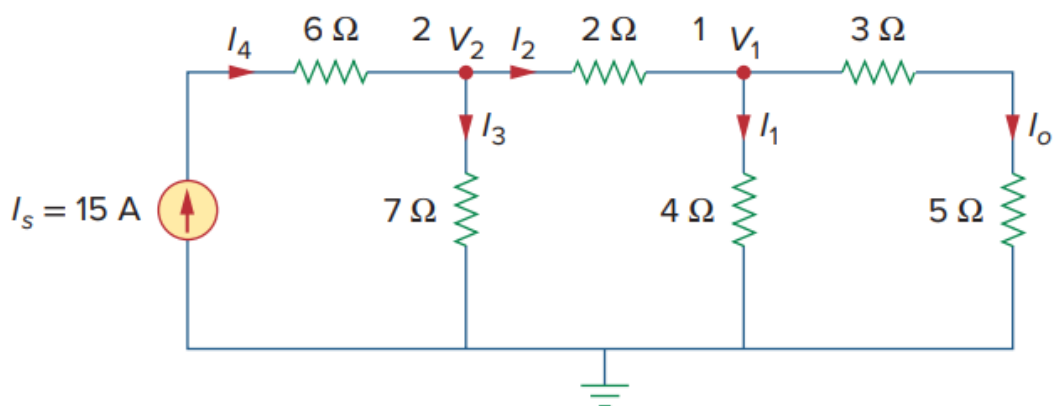


Figure 4.4
For Example 4.2.

Q. Assume $I_o = 1$ A and use linearity to find the actual value of I_o in the circuit of Fig. 4.4.

Solution:

If $I_o = 1$ A, then

$$V_1 = (3 + 5)I_o = 8 \text{ V} \quad \text{and} \quad I_1 = \frac{V_1}{4} = 2 \text{ A}.$$

Applying KCL at node 1 gives

$$I_2 = I_1 + I_o = 3 \text{ A}$$

$$V_2 = V_1 + 2I_2 = 8 + 6 = 14 \text{ V}, \quad I_3 = \frac{V_2}{7} = 2 \text{ A}$$

Applying KCL at node 2 gives

$$I_4 = I_3 + I_2 = 5 \text{ A}$$

Therefore, $I_s = 5$ A. This shows that assuming $I_o = 1$ gives $I_s = 5$ A, the actual source current of 15 A will give $I_o = 3$ A as the actual value.

Superposition principle

1. **The superposition principle** states that the voltage across (or current through) an element in a linear circuit is the algebraic sum of the voltages across (or currents through) that element due to **each independent source** acting alone.
2. The principle of superposition helps us to analyze a linear circuit with more than one independent source by calculating the contribution of each independent source separately. However, to apply the superposition principle, we need to keep two things in mind
 - We consider one independent source at a time while all other independent sources are **turned off**. This implies that we replace every voltage source by 0 V (or a short circuit), and every current source by 0 A (or an open circuit). This way we obtain a simpler and more manageable circuit.
 - **Dependent** sources are **left intact** because they are controlled by circuit variables.
3. **Remark 1.** Analyzing a circuit using superposition has one major disadvantage: It may very likely involve **more work**. If the circuit has three independent sources, we may have to analyze three simpler circuits each providing the contribution due to the respective individual source. However, superposition does help reduce a complex circuit to simpler circuits through replacement of voltage sources by short circuits and of current sources by open circuits.
4. **Remark 2.** Keep in mind that superposition is based on linearity. For this reason, it is not applicable to the effect on power due to each source, because the power absorbed by a resistor depends on the square of the voltage or current. If the power value is needed, the current through (or voltage across) the element must be calculated first using superposition.

Example. Simplest application of Superposition principle

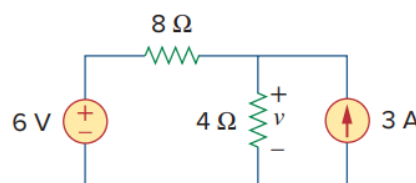
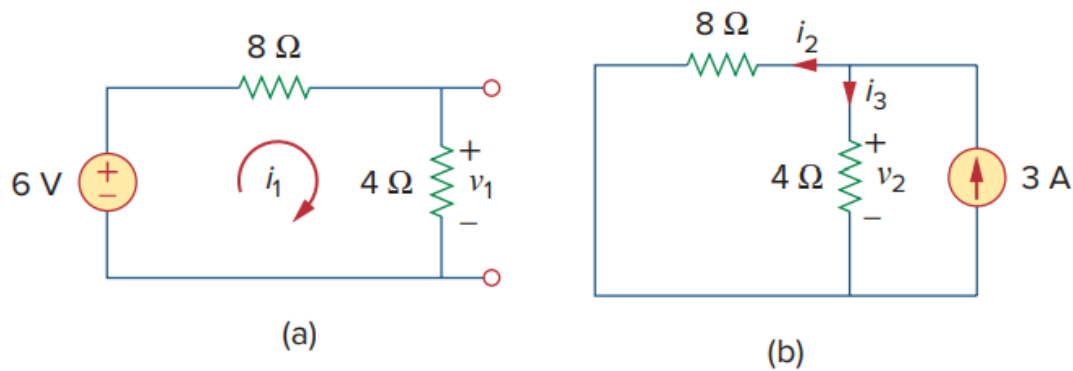


Figure 4.6
For Example 4.3.

Q. Use the superposition theorem to find v in the circuit of Fig. 4.6.

Solution:



Since there are two sources, let

$$v = v_1 + v_2$$

where v_1 and v_2 are the contributions due to the 6-V voltage source and the 3-A current source, respectively. To obtain v_1 , we set the current source to zero, as shown in Fig. 4.7(a). Applying KVL to the loop in Fig. 4.7(a) gives

$$12i_1 - 6 = 0 \quad \Rightarrow \quad i_1 = 0.5 \text{ A}$$

Thus,

$$v_1 = 4i_1 = 2 \text{ V}$$

We may also use voltage division to get v_1 by writing

$$v_1 = \frac{4}{4 + 8}(6) = 2 \text{ V}$$

To get v_2 , we set the voltage source to zero, as in Fig. 4.7(b). Using current division,

$$i_3 = \frac{8}{4 + 8}(3) = 2 \text{ A}$$

Hence,

$$v_2 = 4i_3 = 8 \text{ V}$$

And we find

$$v = v_1 + v_2 = 2 + 8 = 10 \text{ V}$$

Example. Superposition principle applied in the presence of a dependent source.

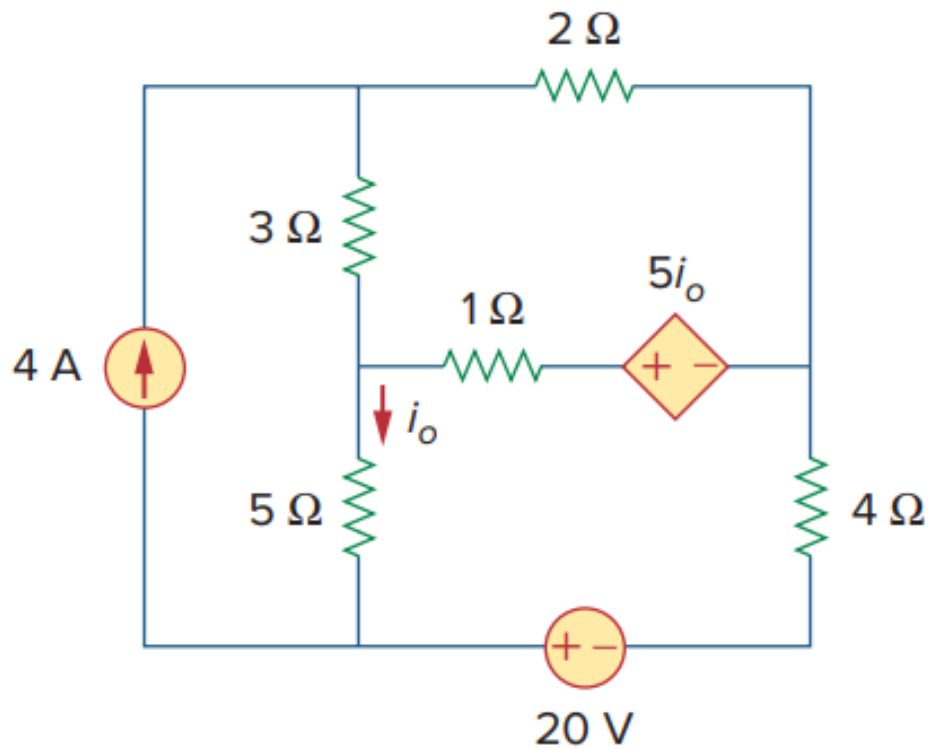


Figure 4.9
For Example 4.4.

Q. Find i_o in the circuit of Fig. 4.9 using superposition.

Solution:

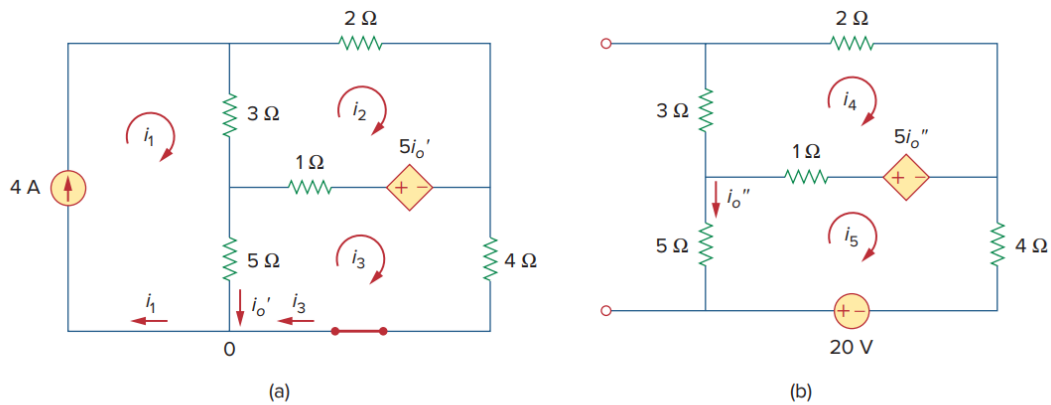


Figure 4.10

For Example 4.4: Applying superposition to (a) obtain i_o' , (b) obtain i_o'' .

The circuit in Fig. 4.9 involves a dependent source, which must be left intact. We let

$$i_o = i_o' + i_o'' \quad (4.4.1)$$

where i_o' and i_o'' are due to the 4-A current source and 20-V voltage source respectively. To obtain i_o' , we turn off the 20-V source so that we have the circuit in Fig. 4.10(a). We apply mesh analysis in order to obtain i_o' . For loop 1,

$$i_1 = 4 \text{ A} \quad (4.4.2)$$

For loop 2,

$$-3i_1 + 6i_2 - i_3 - 5i_o' = 0 \quad (4.4.3)$$

For loop 3,

$$-5i_1 - i_2 + 10i_3 + 5i_o' = 0 \quad (4.4.4)$$

But at node 0,

$$i_3 = i_1 - i_o' = 4 - i_o' \quad (4.4.5)$$

Substituting Eqs. (4.4.2) and (4.4.5) into Eqs. (4.4.3) and (4.4.4) gives two simultaneous equations

$$3i_2 - 2i_o' = 8 \quad (4.4.6)$$

$$i_2 + 5i_o' = 20 \quad (4.4.7)$$

which can be solved to get

$$i_o' = \frac{52}{17} \text{ A} \quad (4.4.8)$$

To obtain i_o'' , we turn off the 4-A current source so that the circuit becomes that shown in Fig. 4.10(b). For loop 4, KVL gives

$$6i_4 - i_5 - 5i_o'' = 0 \quad (4.4.9)$$

and for loop 5,

$$-i_4 + 10i_5 - 20 + 5i_o'' = 0 \quad (4.4.10)$$

But $i_5 = -i_o''$. Substituting this in Eqs. (4.4.9) and (4.4.10) gives

$$6i_4 - 4i_o'' = 0 \quad (4.4.11)$$

$$i_4 + 5i_o'' = -20 \quad (4.4.12)$$

which we solve to get

$$i_o'' = \frac{-60}{17} \text{ A} \quad (4.4.13)$$

Now substituting Eqs. (4.4.8) and (4.4.13) into Eq. (4.4.1) gives

$$i_o = \frac{8}{17} = -0.4706 \text{ A}$$

Example. Three independent sources : Applying all the techniques

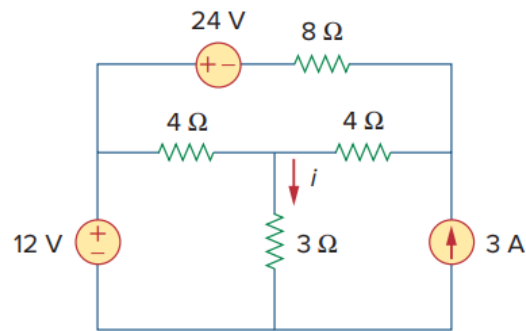


Figure 4.12
For Example 4.5.

Q. For the circuit in Fig. 4.12, use the superposition principle to find i .

Solution:

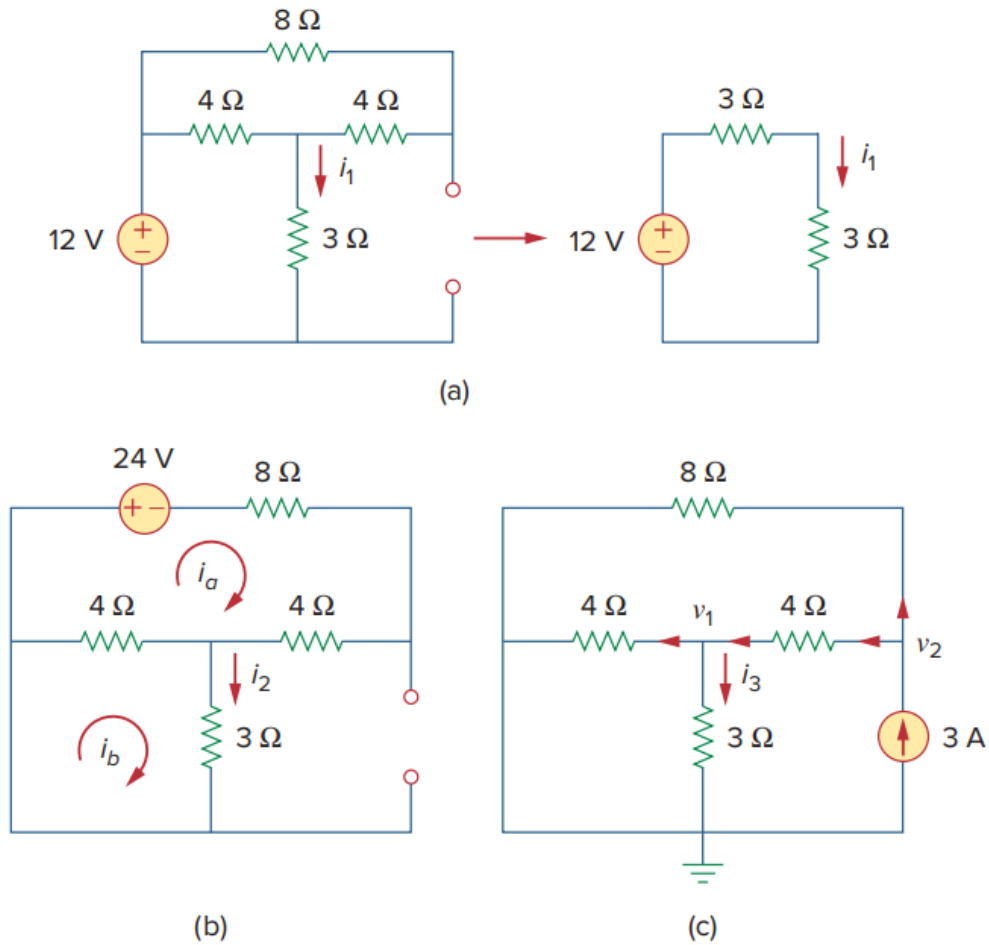


Figure 4.13
For Example 4.5.

In this case, we have three sources. Let

$$i = i_1 + i_2 + i_3$$

where i_1 , i_2 , and i_3 are due to the 12-V, 24-V, and 3-A sources respectively. To get i_1 , consider the circuit in Fig. 4.13(a). Combining 4 Ω (on the right-hand side) in series with 8 Ω gives 12 Ω. The 12 Ω in parallel with 4 Ω gives $12 \times 4/16 = 3 \Omega$. Thus,

$$i_1 = \frac{12}{6} = 2 \text{ A}$$

To get i_2 , consider the circuit in Fig. 4.13(b). Applying mesh analysis gives

$$16i_a - 4i_b + 24 = 0 \quad \Rightarrow \quad 4i_a - i_b = -6 \quad (4.5.1)$$

$$7i_b - 4i_a = 0 \quad \Rightarrow \quad i_a = \frac{7}{4}i_b \quad (4.5.2)$$

Substituting Eq. (4.5.2) into Eq. (4.5.1) gives

$$i_2 = i_b = -1$$

To get i_3 , consider the circuit in Fig. 4.13(c). Using nodal analysis gives

$$3 = \frac{v_2}{8} + \frac{v_2 - v_1}{4} \Rightarrow 24 = 3v_2 - 2v_1 \quad (4.5.3)$$

$$\frac{v_2 - v_1}{4} + \frac{v_1}{4} + \frac{v_1}{3} = v_2 \Rightarrow v_2 = \frac{10}{3}v_1 \quad (4.5.4)$$

Substituting Eq. (4.5.4) into Eq. (4.5.3) leads to $v_1 = 3$ and

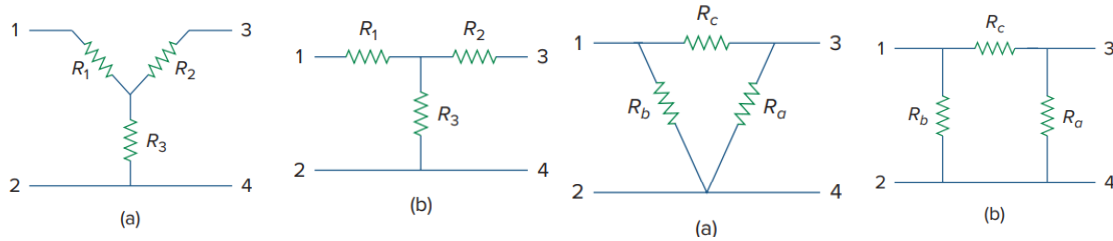
$$i_3 = \frac{v_1}{3} = 1 \text{ A}$$

Thus,

$$i = i_1 + i_2 + i_3 = 2 - 1 + 1 = 2 \text{ A}$$

1.2 Transformation rules for simplifying circuits

1.2.1 Delta-Wye transformation



Derivation of Delta-Wye transformation rule.

To obtain the equivalent resistances in the wye(Y) network, we compare the two networks and make sure that the resistance between each pair of nodes in the Δ (or Π) network is the same as the resistance between the same pair of nodes in the Y (or T) network. For terminals 1 and 2, for instance,

$$R_{12}(\text{Y}) = R_1 + R_3 \quad (1.1)$$

$$R_{12}(\Delta) = R_b \parallel (R_a + R_c) \quad (1.2)$$

Setting $R_{12}(\text{Y}) = R_{12}(\Delta)$ gives:

$$R_{12} = R_1 + R_3 = \frac{R_b(R_a + R_c)}{R_a + R_b + R_c} \quad (1.3)$$

Similarly,

$$R_{13} = R_1 + R_2 = \frac{R_c(R_a + R_b)}{R_a + R_b + R_c} \quad (1.4)$$

$$R_{34} = R_2 + R_3 = \frac{R_a(R_b + R_c)}{R_a + R_b + R_c} \quad (1.5)$$

Subtracting Eq. (5) from Eq. (3), we get:

$$R_1 - R_2 = \frac{R_c(R_b - R_a)}{R_a + R_b + R_c} \quad (1.6)$$

Adding Eqs. (4) and (6) gives:

$$R_1 = \frac{R_b R_c}{R_a + R_b + R_c} \quad (1.7)$$

Subtracting Eq. (6) from Eq. (4) yields:

$$R_2 = \frac{R_c R_a}{R_a + R_b + R_c} \quad (1.8)$$

Subtracting Eq. (7) from Eq. (3), we obtain:

$$R_3 = \frac{R_a R_b}{R_a + R_b + R_c} \quad (1.9)$$

Delta–Wye transformation rule: Each resistor in the Y network is the product of the resistors in the two adjacent Δ branches, divided by the sum of the three Δ resistors.

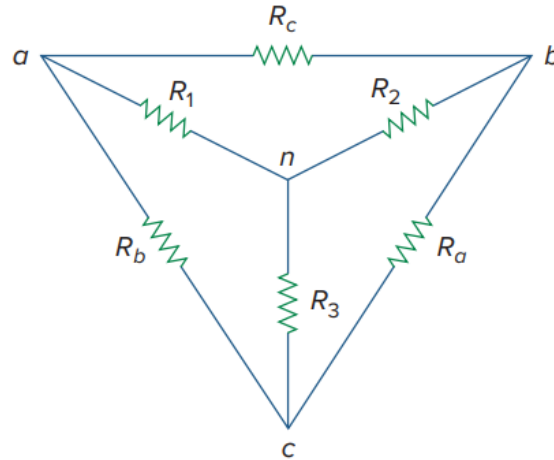


Figure 2.49

Superposition of Y and Δ networks as an aid in transforming one to the other.

1.2.2 Wye-Delta transformation

To obtain the conversion formulas for transforming a wye network to an equivalent delta network, we note from Eqs. (7) to (9) that

$$\begin{aligned} R_1 R_2 + R_2 R_3 + R_3 R_1 &= \frac{R_a R_b R_c (R_a + R_b + R_c)}{(R_a + R_b + R_c)^2} \\ &= \frac{R_a R_b R_c}{R_a + R_b + R_c} \end{aligned} \quad (1.10)$$

Dividing Eq. (10) by each of Eqs. (7) to (9) leads to the following equations:

$$R_a = \frac{R_1 R_2 + R_2 R_3 + R_3 R_1}{R_1} \quad (1.11)$$

$$R_b = \frac{R_1 R_2 + R_2 R_3 + R_3 R_1}{R_2} \quad (1.12)$$

$$R_c = \frac{R_1 R_2 + R_2 R_3 + R_3 R_1}{R_3} \quad (1.13)$$

Wye-Delta transformation rule Each resistor in the Δ network is the sum of all possible products of Y resistors taken two at a time, divided by the opposite Y resistor.

Balanced network The Y and Δ networks are said to be *balanced* when

$$R_1 = R_2 = R_3 = R_Y, \quad R_a = R_b = R_c = R_\Delta \quad (1.14)$$

Under these conditions, conversion formulas become

$$R_Y = \frac{R_\Delta}{3} \quad \text{or} \quad R_\Delta = 3R_Y \quad (1.15)$$

Q. Why $R_Y < R_\Delta$?

To understand why the equivalent resistors in the Y-network are smaller than those in the Δ -network, consider the following operational comparison using the networks shown in Figure 1.1.

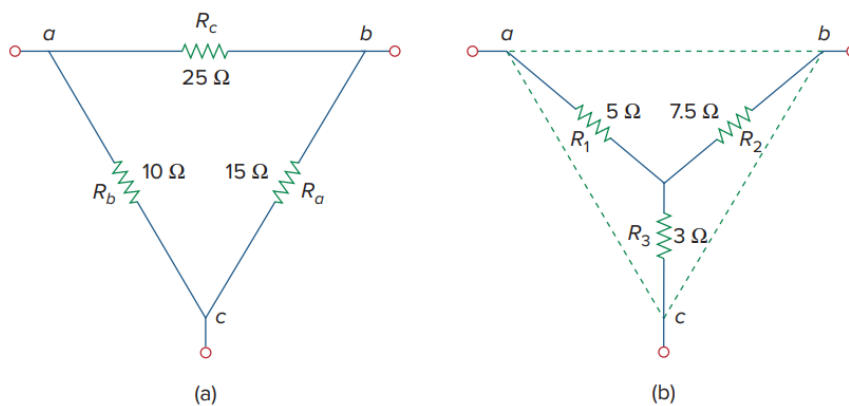


Figure 2.50

For Example 2.14: (a) original Δ network, (b) Y equivalent network.

Figure 1.1: (a) Original Δ network, (b) Y equivalent network.

Suppose a voltage V is applied across terminals a and b in both networks, and the resulting current I is the same. Let us analyze the effective resistance in each configuration.

In the Δ network, current flows between a and b via two possible paths:

- Direct path: through R_c
- Indirect path: from $a \rightarrow c \rightarrow b$, through $R_b + R_a$

These two paths form a parallel connection. The equivalent resistance $R_{\text{eq},\Delta}$ seen from terminals a and b is given by:

$$\frac{1}{R_{\text{eq},\Delta}} = \frac{1}{R_c} + \frac{1}{R_a + R_b}$$

In the Y-network, any current between nodes a and b must pass through *two* resistors in series:

$$R_1 + R_2$$

Hence, the equivalent resistance is simply:

$$R_{\text{eq},Y} = R_1 + R_2$$

To produce the same total current $I = \frac{V}{R_{eq}}$ in both networks with the same applied voltage V , the equivalent resistances must match:

$$R_{eq,\Delta} = R_{eq,Y}$$

Since the Y-network's current path contains two resistors in series, the individual resistors R_1 and R_2 must be smaller than the corresponding Δ resistors R_a , R_b , and R_c to satisfy this condition.

Thus, although the Y-network resembles a “series” shape, its structure leads to a lower equivalent resistance when compared to the Δ network, where parallel paths between nodes allow for smaller overall resistance even though individual resistors may be larger.

Example Obtain the equivalent resistance R_{ab} for the circuit in Fig. 2.52 and use it to find current i .

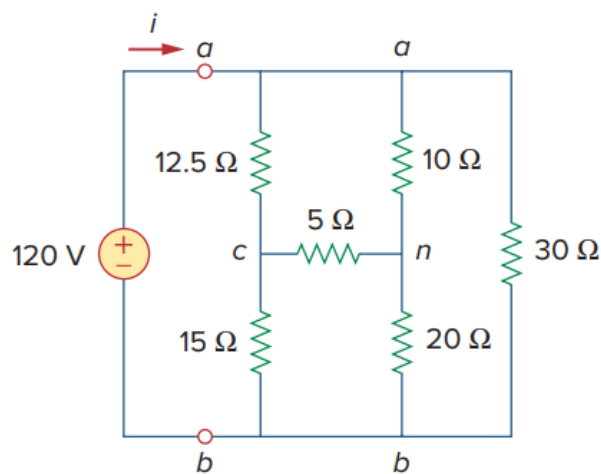
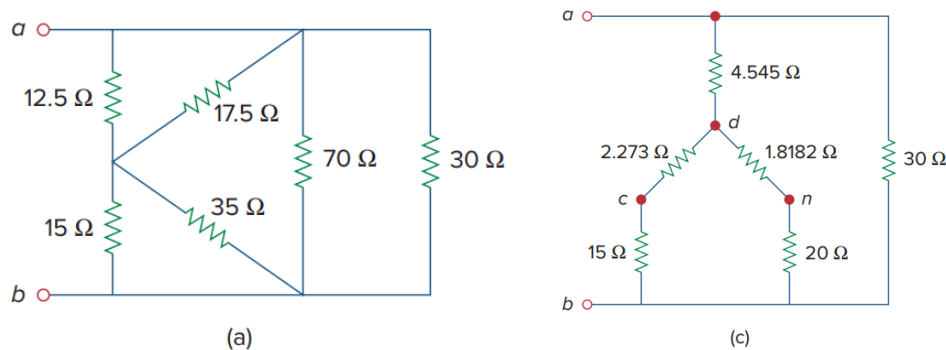


Figure 2.52
For Example 2.15.

There are two ways to simplify this circuit:



1. Like (a), Y network (acb) can be converted to a delta network. Equivalent resistances can be calculated as

$$R_{ab} = \{(12.5\Omega \parallel 17.5\Omega) + (15\Omega \parallel 35\Omega)\} \parallel (70\Omega \parallel 30\Omega)$$

2. Like (b), a delta network acn can be converted to a wye network.

1.2.3 Source Transformation

1. **Motivation.** Transformation rules for simplifying circuits such as delta-wye transformation rests on the concept of **equivalence**. **Equivalent circuit** is one whose $v - i$ characteristics are identical with the original one.
2. The two circuits in Fig. 4.15 are equivalent—provided they have the same voltage-current relation at terminals a - b .

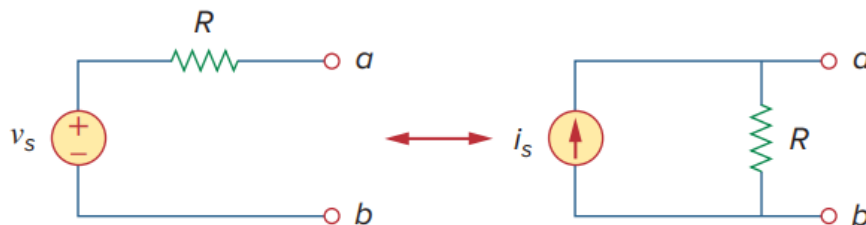


Figure 4.15

Transformation of independent sources.

It is easy to show that they are indeed equivalent. If the sources are turned off, the equivalent resistance at terminals a - b in both circuits is R . Also, when terminals a - b are short-circuited, the short-circuit current flowing from a to b is $i_{sc} = v_s/R$ in the circuit on the left-hand side and $i_{sc} = i_s$ for the circuit on the right-hand side. Thus, $v_s/R = i_s$ in order for the two circuits to be equivalent. Hence, source transformation requires that

$$v_s = i_s R \quad \text{or} \quad i_s = \frac{v_s}{R} \quad (4.5)$$

A source transformation is the process of replacing a voltage source v_s in series with a resistor R by a current source i_s in parallel with a resistor R , or vice versa.

3. Source transformation also applies to dependent sources, provided we carefully handle the dependent variable.

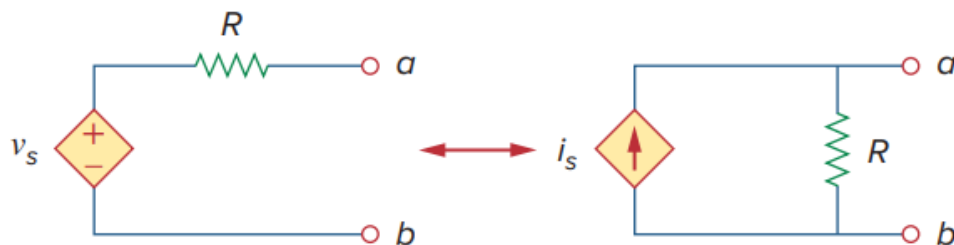


Figure 4.16

Transformation of dependent sources.

As shown in Fig. 4.16, a dependent voltage source in series with a resistor can be transformed to a dependent current source in parallel with the resistor or vice versa where we make sure that Eq. (4.5) is satisfied.

4. **Remarks.** Like the wye-delta transformation, a source transformation does not affect the remaining part of the circuit. When applicable, source transformation is a powerful tool that allows circuit manipulations to ease circuit analysis. However, we should keep the following points in mind when dealing with source transformation.
- (a) Note from Fig. 4.15 (or Fig. 4.16) that the arrow of the current source is directed toward the positive terminal of the voltage source.
 - (b) Note from Eq. (4.5) that source transformation is not possible when $R = 0$, which is the case with an ideal voltage source. However, for a practical, nonideal voltage source, $R \neq 0$. Similarly, an ideal current source with $R = \infty$ cannot be replaced by a finite voltage source.

Example. Simplest application of source transformation

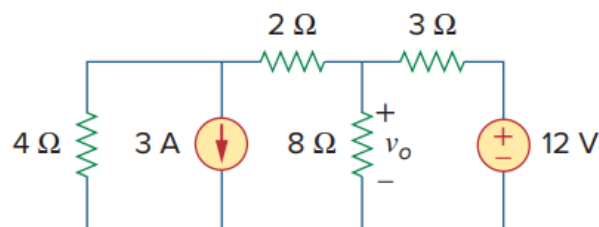


Figure 4.17
For Example 4.6.

Use source transformation to find v_o in the circuit of Fig. 4.17.

Solution:

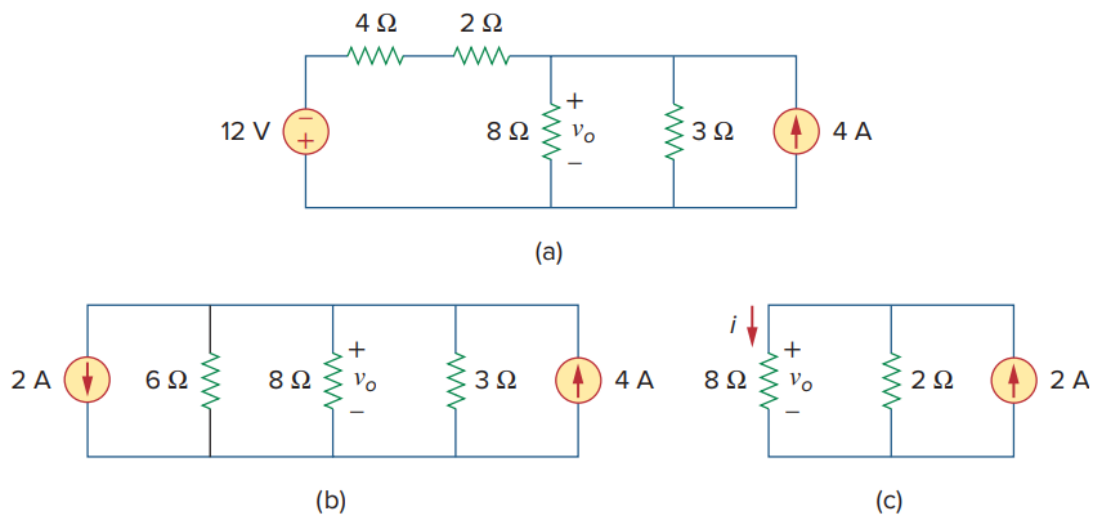


Figure 4.18
For Example 4.6.

We first transform the current and voltage sources to obtain the circuit in Fig. 4.18(a). Combining the $4\text{-}\Omega$ and $2\text{-}\Omega$ resistors in series and transforming the 12-V voltage source gives us Fig. 4.18(b). We now combine the $3\text{-}\Omega$ and $6\text{-}\Omega$ resistors in parallel to get $2\text{-}\Omega$. We also combine the 2-A and 4-A current sources to get a 2-A source. Thus, by repeatedly applying source transformations, we obtain the circuit in Fig. 4.18(c).

We use current division in Fig. 4.18(c) to get

$$i = \frac{2}{2+8}(2) = 0.4\text{ A}$$

and

$$v_o = 8i = 8(0.4) = 3.2\text{ V}$$

Alternatively, since the $8\text{-}\Omega$ and $2\text{-}\Omega$ resistors in Fig. 4.18(c) are in parallel, they have the same voltage v_o across them. Hence,

$$v_o = (8 \parallel 2)(2\text{ A}) = \frac{8 \times 2}{10}(2) = 3.2\text{ V}$$

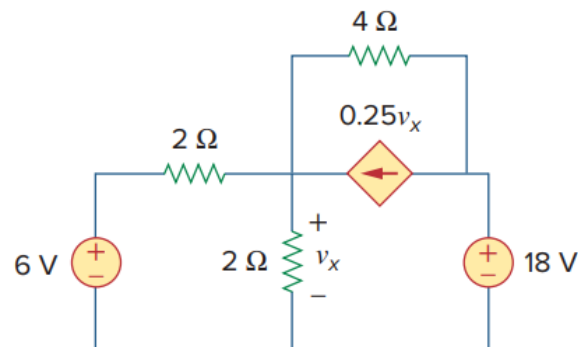
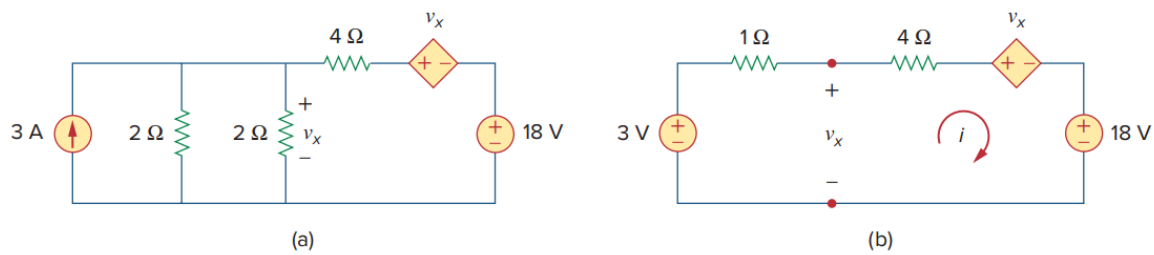
Example. Source transformation applied to a dependent source

Figure 4.20
For Example 4.7.

Find v_x in Fig. 4.20 using source transformation.

**Figure 4.21**

For Example 4.7: Applying source transformation to the circuit in Fig. 4.20.

Solution:

The circuit in Fig. 4.20 involves a voltage-controlled dependent current source. We transform this dependent current source as well as the 6-V independent voltage source as shown in Fig. 4.21(a). The 18-V voltage source is not transformed because it is not connected in series with any resistor. The two 2-Ω resistors in parallel combine to give a 1-Ω resistor, which is in parallel with the 3-A current source. The current source is transformed to a voltage source as shown in Fig. 4.21(b). Notice that the terminals for v_x are intact. Applying KVL around the loop in Fig. 4.21(b) gives

$$-3 + 5i + v_x + 18 = 0 \quad (4.7.1)$$

Applying KVL to the loop containing only the 3-V voltage source, the 1-Ω resistor, and v_x yields

$$-3 + 1i + v_x = 0 \quad \Rightarrow \quad v_x = 3 - i \quad (4.7.2)$$

Substituting this into Eq. (4.7.1), we obtain

$$15 + 5i + 3 - i = 0 \quad \Rightarrow \quad i = -4.5 \text{ A}$$

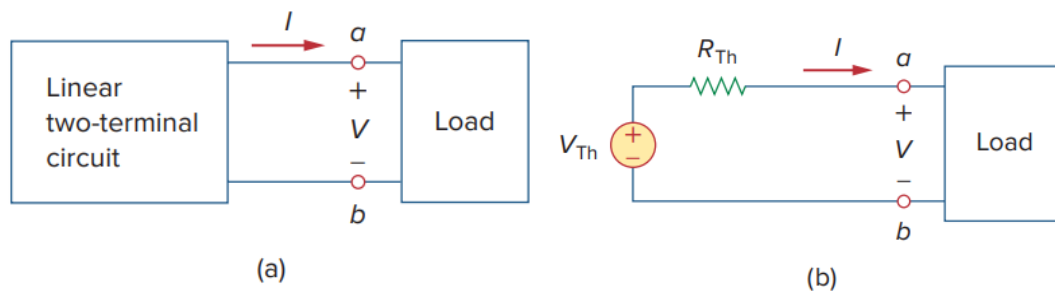
Alternatively, we may apply KVL to the loop containing v_x , the 4-Ω resistor, the voltage-controlled dependent voltage source, and the 18-V voltage source in Fig. 4.21(b). We obtain

$$-v_x + 4i + v_x + 18 = 0 \quad \Rightarrow \quad i = -4.5 \text{ A}$$

Thus, $v_x = 3 - i = 7.5 \text{ V}$.

1.2.4 Thevenin theorem

1. **Motivation.** It often occurs in practice that a particular element in a circuit is variable (usually called the load) while other elements are fixed. As a typical example, a household outlet terminal may be connected to different appliances constituting a variable load. Each time the variable element is changed, the entire circuit has to be analyzed all over again. To avoid this problem, Thevenin's theorem provides a technique by which the fixed part of the circuit is replaced by an equivalent circuit.
2. According to Thevenin's theorem, the linear circuit in Fig. 4.23(a) can be replaced by that in Fig. 4.23(b). The circuit to the left of the terminals a-b in Fig. 4.23(b) is known as the Thevenin equivalent circuit;



Thevenin's theorem states that a linear two-terminal circuit can be replaced by an equivalent circuit consisting of a voltage source V_{Th} in series with a resistor R_{Th} , where V_{Th} is the open-circuit voltage at the terminals and R_{Th} is the input or equivalent resistance at the terminals when the independent sources are turned off.

3. **How to find the Thevenin equivalent voltage.** V_{Th} and resistance R_{Th} . To do so, suppose the two circuits in Fig. 4.23 are equivalent. Two circuits are said to be *equivalent* if they have the same voltage-current relation at their terminals. Let us find out what will make the two circuits in Fig. 4.23 equivalent.

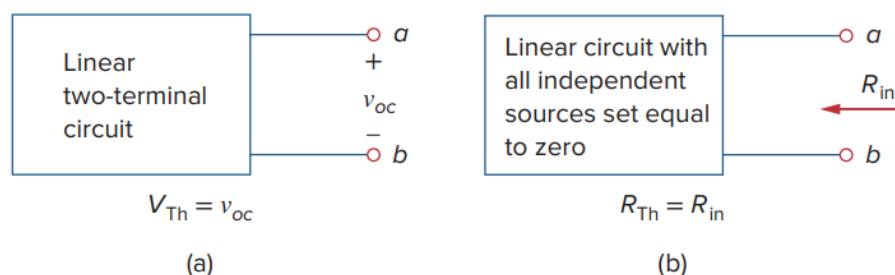


Figure 4.24
Finding V_{Th} and R_{Th} .

If the terminals a - b are made open-circuited (by removing the load), no current flows, so that the open-circuit voltage across the terminals a - b in Fig. 4.23(a) must be equal to the voltage source V_{Th} in Fig. 4.23(b), since the two circuits are equivalent. Thus, V_{Th} is the open-circuit voltage across the terminals as shown in Fig. 4.24(a); that is,

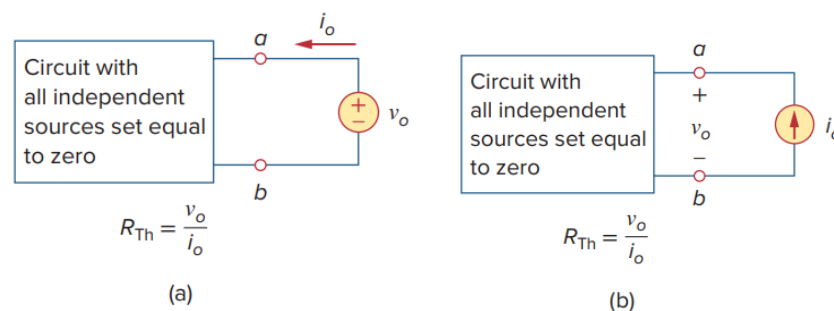
$$V_{Th} = v_{oc} \quad (4.6)$$

Again, with the load disconnected and terminals a - b open-circuited, we turn off all independent sources. The input resistance (or equivalent resistance) of the dead circuit at the terminals a - b in Fig. 4.23(a) must be equal to R_{Th} in Fig. 4.23(b) because the two circuits are equivalent. Thus, R_{Th} is the input resistance at the terminals when the independent sources are turned off, as shown in Fig. 4.24(b); that is,

$$R_{Th} = R_{in} \quad (4.7)$$

To apply this idea in finding the Thevenin resistance R_{Th} , we need to consider two cases.

- **CASE 1** If the network has no dependent sources, we turn off all independent sources. R_{Th} is the input resistance of the network looking between terminals a and b , as shown in Fig. 4.24(b).



- **CASE 2** If the network has dependent sources, we turn off all independent sources. As with superposition, dependent sources are not to be turned off because they are controlled by circuit variables. We apply a voltage source v_o at terminals a and b and determine the resulting current i_o . Then $R_{Th} = v_o/i_o$, as shown in Fig. 4.25(a). Alternatively, we may insert a current source i_o at terminals a - b as shown in Fig. 4.25(b) and find the terminal voltage v_o . Again $R_{Th} = v_o/i_o$. Either of the two approaches will give the same result. In either approach we may assume any value of v_o and i_o . For example, we may use $v_o = 1$ V or $i_o = 1$ A, or even use unspecified values of v_o or i_o .

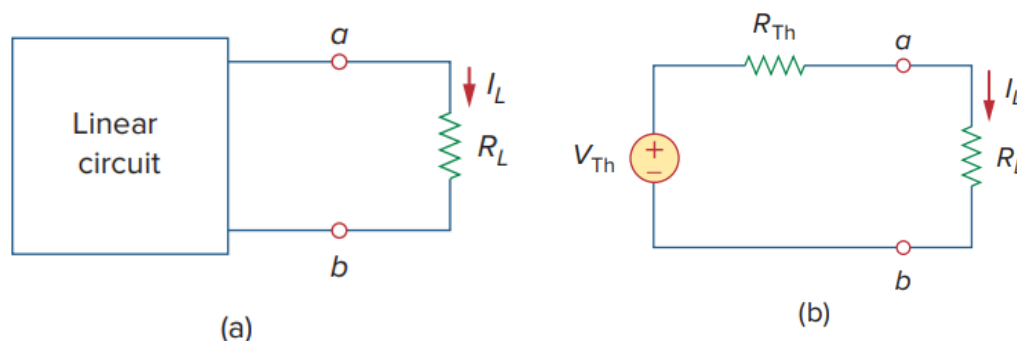


Figure 4.26

4. As mentioned earlier, a linear circuit with a variable load can be replaced by the Thevenin equivalent, exclusive of the load. The equivalent network behaves the same way externally as the original circuit. Consider a linear circuit terminated by a load R_L , as shown in Fig. 4.26(a).

The current I_L through the load and the voltage V_L across the load are easily determined once the Thevenin equivalent of the circuit at the load's terminals is obtained, as shown in Fig. 4.26(b). From Fig. 4.26(b), we obtain

$$I_L = \frac{V_{Th}}{R_{Th} + R_L} \quad (4.8a)$$

$$V_L = R_L I_L = \frac{R_L}{R_{Th} + R_L} V_{Th} \quad (4.8b)$$

Note from Fig. 4.26(b) that the Thevenin equivalent is a simple voltage divider, yielding V_L by mere inspection.

Example. Finding Thevenin Equivalent circuit : Case 1 - No dependent sources

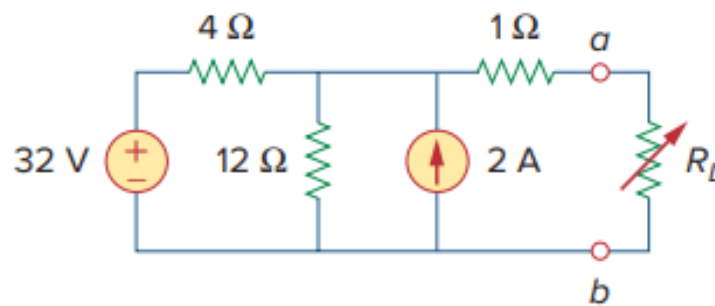
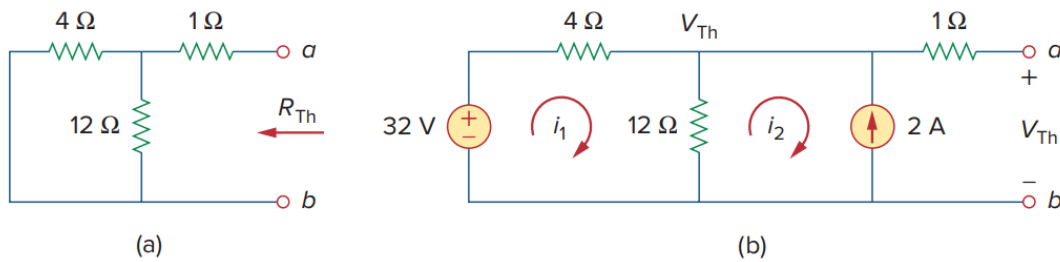


Figure 4.27

Find the Thevenin equivalent circuit of the circuit shown in Fig. 4.27, to the left of the terminals a - b . Then find the current through $R_L = 6, 16$, and 36Ω .

Solution:

We find R_{Th} by turning off the 32-V voltage source (replacing it with a short circuit)

**Figure 4.28**

For Example 4.8: (a) finding R_{Th} , (b) finding V_{Th} .

and the 2-A current source (replacing it with an open circuit). The circuit becomes what is shown in Fig. 4.28(a). Thus,

$$R_{Th} = 4 \parallel 12 + 1 = \frac{4 \times 12}{16} + 1 = 4 \Omega$$

To find V_{Th} , consider the circuit in Fig. 4.28(b). Applying mesh analysis to the two loops, we obtain

$$-32 + 4i_1 + 12(i_1 - i_2) = 0, \quad i_2 = -2 \text{ A}$$

Solving for i_1 , we get $i_1 = 0.5 \text{ A}$. Thus,

$$V_{Th} = 12(i_1 - i_2) = 12(0.5 + 2.0) = 30 \text{ V}$$

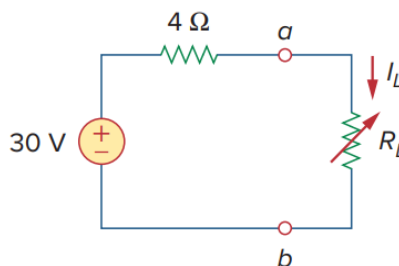
Alternatively, it is even easier to use nodal analysis. We ignore the 1- Ω resistor since no current flows through it. At the top node, KCL gives

$$\frac{32 - V_{Th}}{4} + 2 = \frac{V_{Th}}{12}$$

or

$$96 - 3V_{Th} + 24 = V_{Th} \Rightarrow V_{Th} = 30 \text{ V}$$

as obtained before. We could also use source transformation to find V_{Th} .

**Figure 4.29**

The Thevenin equivalent circuit for Example 4.8.

The Thevenin equivalent circuit is shown in Fig. 4.29. The current through R_L is

$$I_L = \frac{V_{Th}}{R_{Th} + R_L} = \frac{30}{4 + R_L}$$

When $R_L = 6$, $I_L = \frac{30}{10} = 3 \text{ A}$; When $R_L = 16$, $I_L = \frac{30}{20} = 1.5 \text{ A}$; When $R_L = 36$, $I_L = \frac{30}{40} = 0.75 \text{ A}$

Example. Finding Thevenin Equivalent circuit : Case 2 - With dependent sources

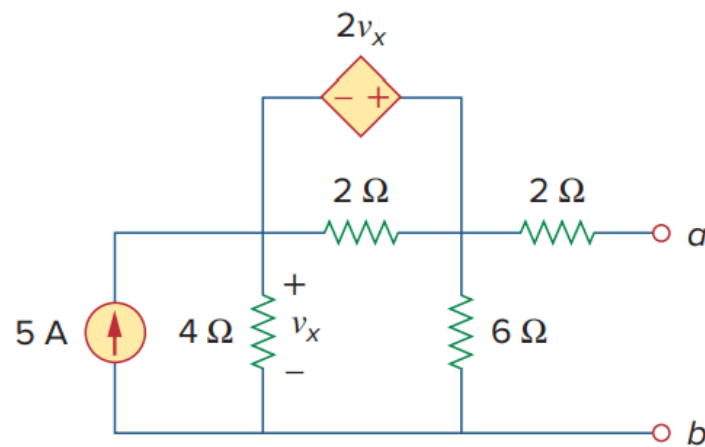


Figure 4.31
For Example 4.9.

Q. Find the Thevenin equivalent of the circuit in Fig. 4.31 at terminals $a - b$.

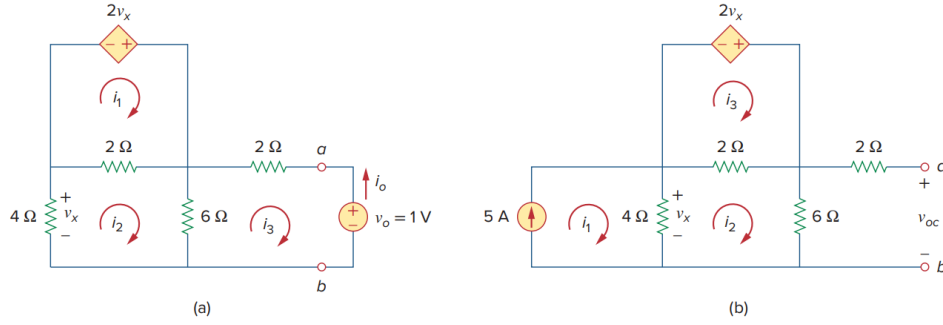
Solution:

Figure 4.32
Finding R_{Th} and V_{Th} for Example 4.9.

This circuit contains a dependent source, unlike the circuit in the previous example. To find R_{Th} , we set the independent source equal to zero but leave the dependent source alone. Because of the presence of the dependent source, however, we excite the network with a voltage source v_o connected to the terminals as indicated in Fig. 4.32(a). We may set $v_o = 1\text{ V}$ to ease calculation, since the circuit is linear. Our goal is to find the current i_o through the terminals, and then obtain $R_{Th} = 1/i_o$. (Alternatively, we may insert a 1-A current source, find the corresponding voltage v_o , and obtain $R_{Th} = v_o/1$.)

Applying mesh analysis to loop 1 in the circuit of Fig. 4.32(a) results in

$$-2v_x + 2(i_1 - i_2) = 0 \quad \text{or} \quad v_x = i_1 - i_2$$

But $-4i_2 = v_x = i_1 - i_2$; hence,

$$i_1 = -3i_2 \quad (4.9.1)$$

For loops 2 and 3, applying KVL produces

$$4i_2 + 2(i_2 - i_1) + 6(i_2 - i_3) = 0 \quad (4.9.2)$$

$$6(i_3 - i_2) + 2i_3 + 1 = 0 \quad (4.9.3)$$

Solving these equations gives

$$i_3 = -\frac{1}{6}\text{ A}$$

But $i_o = -i_3 = \frac{1}{6}\text{ A}$. Hence,

$$R_{Th} = \frac{1\text{ V}}{i_o} = 6\ \Omega$$

To get V_{Th} , we find v_{oc} in the circuit of Fig. 4.32(b). Applying mesh analysis, we get

$$i_1 = 5 \quad (4.9.4)$$

$$-2v_x + 2(i_3 - i_2) = 0 \quad \Rightarrow \quad v_x = i_3 - i_2 \quad (4.9.5)$$

$$4(i_2 - i_1) + 2(i_2 - i_3) + 6i_2 = 0$$

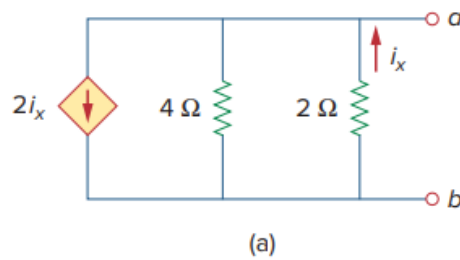
or

$$12i_2 - 4i_1 - 2i_3 = 0 \quad (4.9.6)$$

But $4(i_1 - i_2) = v_x$. Solving these equations leads to $i_2 = 10/3$. Hence,

$$V_{Th} = v_{oc} = 6i_2 = 20\text{ V}$$

Example. Finding Thevenin Equivalent circuit : Case 3 - Only dependent sources



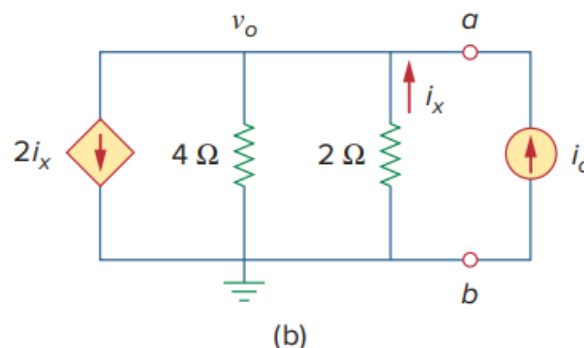
Determine the Thevenin equivalent of the circuit in Fig. 4.35(a) at terminals a-b.

Solution:

1. **Define.** The problem is clearly defined; we are to determine the Thevenin equivalent of the circuit shown in Fig. 4.35(a).
2. **Present.** The circuit contains a $2\text{-}\Omega$ resistor in parallel with a $4\text{-}\Omega$ resistor. These are, in turn, in parallel with a dependent current source. It is important to note that there are no independent sources.
3. **Alternative.** The first thing to consider is that, since we have no independent sources in this circuit, we must excite the circuit externally. In addition, when you have no independent sources the value for V_{Th} will be equal to zero so you will only have to find R_{Th} .

The simplest approach is to excite the circuit with either a 1-V voltage source or a 1-A current source. Because we will end up with an equivalent resistance (either positive or negative), I prefer to use the current source and nodal analysis which will yield a voltage at the output terminals equal to the resistance (with 1 A flowing in, v_o is equal to 1 times the equivalent resistance).

As an alternative, the circuit could also be excited by a 1-V voltage source and mesh analysis could be used to find the equivalent resistance.



4. **Attempt.** We start by writing the nodal equation at a in Fig. 4.35(b) assuming $i_o = 1\text{ A}$.

$$2i_x + \frac{v_o}{4} + \frac{v_o}{2} + (-1) = 0 \quad (4.10.1)$$

Given that we have two unknowns and only one equation, we will need a constraint equation.

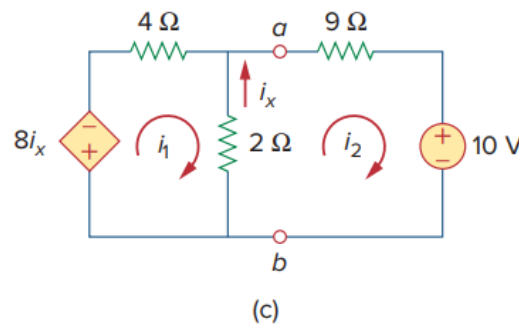
$$i_x = \frac{0 - v_o}{2} = -\frac{v_o}{2} \quad (4.10.2)$$

Substituting Eq. (4.10.2) into Eq. (4.10.1) yields:

$$\begin{aligned} 2(-v_o/2) + \frac{v_o}{4} + \frac{v_o}{2} + (-1) &= 0 \\ = \left(-1 + \frac{1}{4} + \frac{1}{2}\right)v_o - 1 &= 0 \Rightarrow v_o = -4 \text{ V} \end{aligned}$$

Since $v_o = 1 \times R_{Th}$, then $R_{Th} = v_o/1 = -4 \Omega$

The negative value of the resistance tells us that, according to the passive sign convention, the circuit in Fig. 4.35(a) is supplying power. Of course, the resistors in Fig. 4.35(a) cannot supply power (they absorb power); it is the dependent source that supplies the power. This is an example of how a dependent source and resistors could be used to simulate negative resistance.



5. **Evaluate.** First of all, we note that the answer has a negative value. We know this is not possible in a passive circuit, but in this circuit we do have an active device (the dependent current source). Thus, the equivalent circuit is essentially an active circuit that can supply power.

Now we must evaluate the solution. The best way to do this is to perform a check, using a different approach, and see if we obtain the same solution. Let us try connecting a 9- Ω resistor in series with a 10-V voltage source across the output terminals of the original circuit and then the Thevenin equivalent. To make the circuit easier to solve, we can take and change the parallel current source and 4- Ω resistor to a series voltage source and 4- Ω resistor by using source transformation. This, with the new load, gives us the circuit shown in Fig. 4.35(c).

We can now write two mesh equations.

$$\begin{aligned} 8i_x + 4i_1 + 2(i_1 - i_2) &= 0 \\ 2(i_2 - i_1) + 9i_2 + 10 &= 0 \end{aligned}$$

Note, we only have two equations but have three unknowns, so we need a constraint equation. We can use

$$i_x = i_2 - i_1$$

This leads to a new equation for loop 1. Simplifying leads to:

$$(4 + 2 - 8)i_1 + (-2 + 8)i_2 = 0 \quad \text{or} \quad -2i_1 + 6i_2 = 0 \quad \Rightarrow \quad i_1 = 3i_2$$

$$\text{Substitute into second: } -2i_1 + 11i_2 = -10 \quad \Rightarrow \quad -6i_2 + 11i_2 = -10 \quad \Rightarrow \quad i_2 = -2 \text{ A}$$

Using the Thevenin equivalent is quite easy since we have only one loop, as shown in Fig. 4.35(d):

$$-4i + 9i + 10 = 0 \quad \Rightarrow \quad i = -\frac{10}{5} = -2 \text{ A}$$

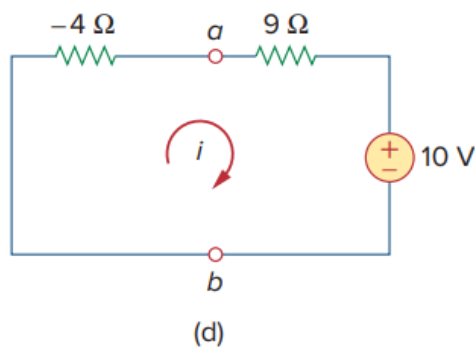


Figure 4.35

6. **Satisfactory?** Clearly we have found the value of the equivalent circuit as required by the problem statement. Checking does validate that solution (we compared the answer we obtained by using the equivalent circuit with one obtained by using the load with the original circuit). We can present all this as a solution to the problem.

1.2.5 Maximum Power Transfer theorem

1. **Motivation.** In many practical situations, a circuit is designed to provide power to a load. There are applications in areas such as communications where it is desirable to maximize the power delivered to a load. We now address the problem of delivering the maximum power to a load when given a system with known internal losses. It should be noted that this will result in significant internal losses greater than or equal to the power delivered to the load.

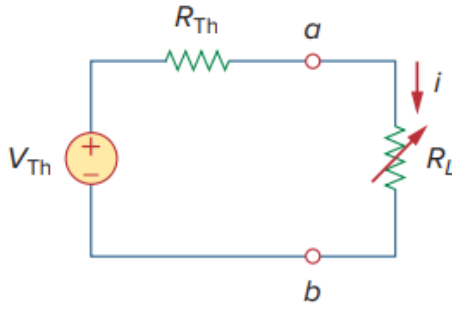


Figure 4.48
The circuit used for maximum power transfer.

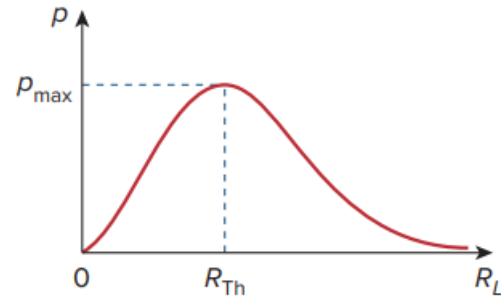


Figure 4.49
Power delivered to the load as a function of R_L .

2. The Thevenin equivalent is useful in finding the maximum power a linear circuit can deliver to a load. We assume that we can adjust the load resistance R_L . If the entire circuit is replaced by its Thevenin equivalent except for the load, as shown in Fig. 4.48, the power delivered to the load is

$$p = i^2 R_L = \left(\frac{V_{Th}}{R_{Th} + R_L} \right)^2 R_L \quad (4.21)$$

For a given circuit, V_{Th} and R_{Th} are fixed. By varying the load resistance R_L , the power delivered to the load varies as sketched in Fig. 4.49. We notice from Fig. 4.49 that the power is small for small or large values of R_L but maximum for some value of R_L between 0 and ∞ . We now want to show that this maximum power occurs when R_L is equal to R_{Th} . This is known as the *maximum power theorem*.

Maximum Power Theorem

Maximum power is transferred to the load when the load resistance equals the Thevenin resistance as seen from the load ($R_L = R_{Th}$).

3. **Proof.** To prove the maximum power transfer theorem, we differentiate p in Eq. (4.21) with respect to R_L and set the result equal to zero. We obtain

$$\begin{aligned} \frac{dp}{dR_L} &= V_{Th}^2 \left[\frac{(R_{Th} + R_L)^2 - 2R_L(R_{Th} + R_L)}{(R_{Th} + R_L)^4} \right] \\ &= V_{Th}^2 \left[\frac{(R_{Th} + R_L - 2R_L)}{(R_{Th} + R_L)^3} \right] = 0 \end{aligned}$$

This implies that

$$0 = (R_{Th} + R_L - 2R_L) = (R_{Th} - R_L) \quad (4.22)$$

which yields

$$R_L = R_{Th} \quad (4.23)$$

showing that the maximum power transfer takes place when the load resistance R_L equals the Thevenin resistance R_{Th} . We can readily confirm that Eq. (4.23) gives the maximum power by showing that $d^2p/dR_L^2 < 0$.

The maximum power transferred is obtained by substituting Eq. (4.23) into Eq. (4.21), for

$$p_{\max} = \frac{V_{Th}^2}{4R_{Th}} \quad (4.24)$$

Equation (4.24) applies only when $R_L = R_{Th}$. When $R_L \neq R_{Th}$, we compute the power delivered to the load using Eq. (4.21).

Example. Application of Maximum power theorem

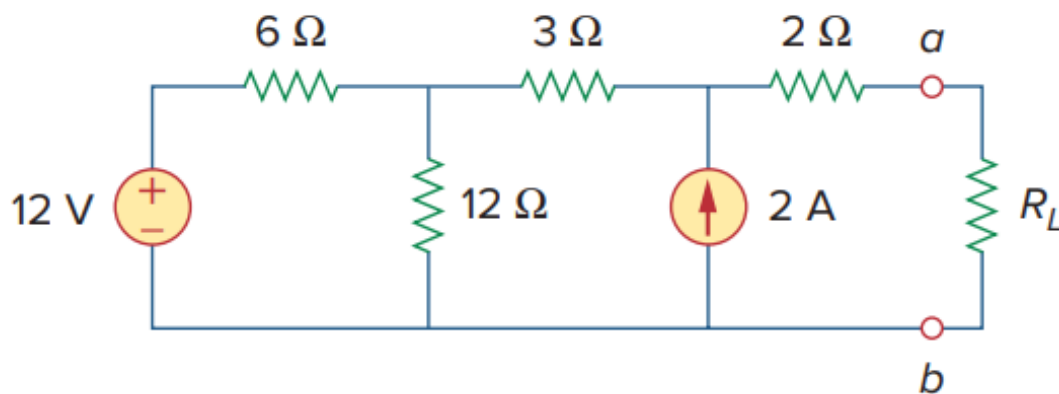


Figure 4.50
For Example 4.13.

Find the value of R_L for maximum power transfer in the circuit of Fig. 4.50. Find the maximum power.

Solution: Solving for i_1 , we get $i_1 = -2/3$. Applying KVL around the outer loop to get

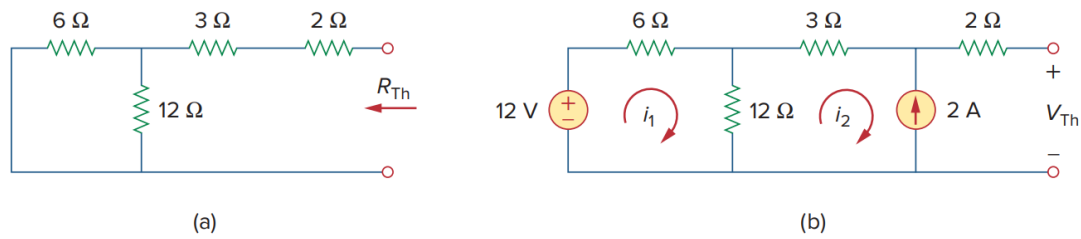


Figure 4.51

For Example 4.13: (a) finding R_{Th} , (b) finding V_{Th} .

V_{Th} across terminals a – b , we obtain

$$-12 + 6i_1 + 3i_2 + 2(0) + V_{Th} = 0 \quad \Rightarrow \quad V_{Th} = 22\text{ V}$$

For maximum power transfer,

$$R_L = R_{Th} = 9\ \Omega$$

and the maximum power is

$$p_{\max} = \frac{V_{Th}^2}{4R_L} = \frac{22^2}{4 \times 9} = 13.44\text{ W}$$

1.2.6 Norton theorem

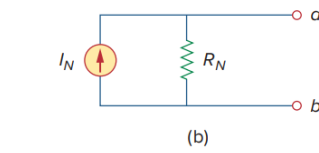
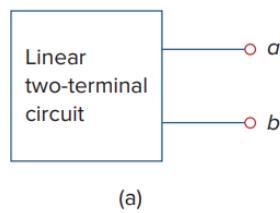


Figure 4.37
(a) Original circuit, (b) Norton equivalent circuit.

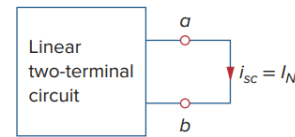


Figure 4.38
Finding Norton current I_N .

1. **Norton's theorem** states that a linear two-terminal circuit can be replaced by an equivalent circuit consisting of a current source I_N in parallel with a resistor R_N , where I_N is the **short-circuit current** through the terminals and R_N is the input or equivalent resistance at the terminals when the independent sources are turned off.
2. **How to find the Norton current I_N .** To find the Norton current I_N , we determine the short-circuit current flowing from terminal a to b in both circuits in Fig. 4.37. It is evident that the short-circuit current in Fig. 4.37(b) is I_N . This must be the same short-circuit current from terminal a to b in Fig. 4.37(a), since the two circuits are equivalent. Thus

$$I_N = i_{sc}$$

3. **The close relationship between Thevenin and Norton theorem.** By source transformation,

$$I_N = \frac{V_{Th}}{R_{Th}}$$

$$R_N = R_{Th}$$

To determine the Thevenin or Norton equivalent circuit requires that we find:

- The open-circuit voltage v_{oc} across terminals a and b .
- The short-circuit current i_{sc} at terminals a and b .
- The equivalent or input resistance R_{in} at terminals a and b when all independent sources are turned off.

We can calculate any two of the three using the method that takes the least effort and use them to get the third using Ohm's law. Also, since

$$V_{Th} = v_{oc} \quad (4.12a)$$

$$I_N = i_{sc} \quad (4.12b)$$

$$R_{Th} = \frac{v_{oc}}{i_{sc}} = R_N \quad (4.12c)$$

the open-circuit and short-circuit tests are sufficient to find any Thevenin or Norton equivalent, of a circuit which contains at least one independent source.

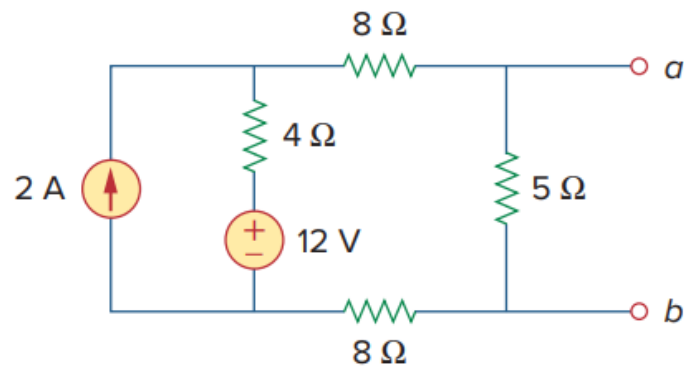
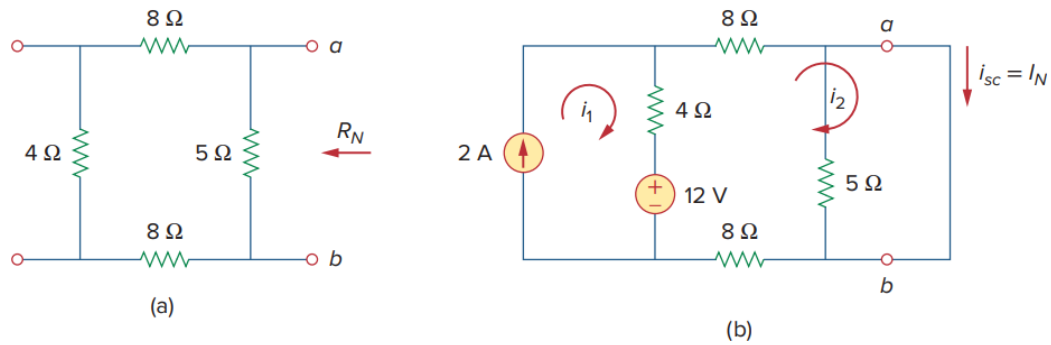
Example. Finding Norton current

Figure 4.39
For Example 4.11.

Find the Norton equivalent circuit of the circuit in Fig. 4.39 at terminals $a - b$.

**Solution:**

We find R_N in the same way we find R_{Th} in the Thevenin equivalent circuit. Set the independent sources equal to zero. This leads to the circuit in Fig. 4.40(a), from which we find R_N . Thus,

$$R_N = 5 \parallel (8 + 4 + 8) = 5 \parallel 20 = \frac{20 \times 5}{25} = 4 \Omega$$

To find I_N , we short-circuit terminals a and b , as shown in Fig. 4.40(b). We ignore the $5\text{-}\Omega$ resistor because it has been short-circuited. Applying mesh analysis, we obtain

$$i_1 = 2 \text{ A}, \quad 20i_2 - 4i_1 - 12 = 0$$

From these equations, we obtain

$$i_2 = 1 \text{ A} = i_{sc} = I_N$$

Example. Finding the Norton equivalent in the presence of a dependent source

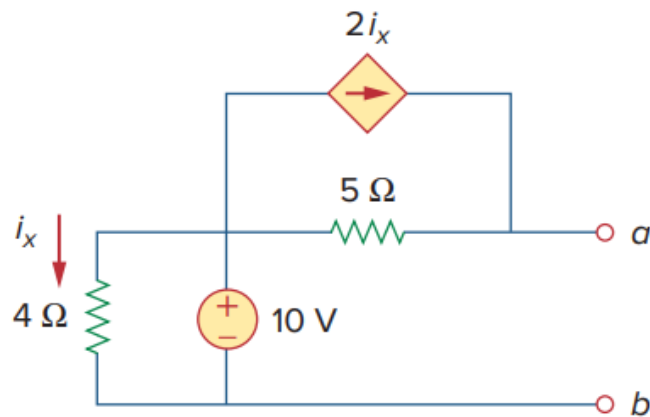
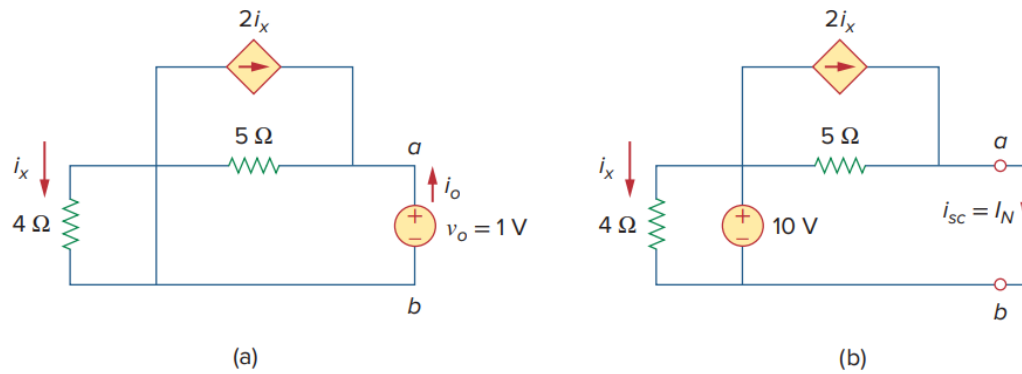


Figure 4.43
For Example 4.12.

Using Norton's theorem, find R_N and I_N of the circuit in Fig. 4.43 at terminals $a - b$.

**Figure 4.44**For Example 4.12: (a) finding R_N , (b) finding I_N .**Solution:**

To find R_N , we set the independent voltage source equal to zero and connect a voltage source of $v_o = 1\text{ V}$ (or any unspecified voltage v_o) to the terminals. We obtain the circuit in Fig. 4.44(a). We ignore the $4\text{-}\Omega$ resistor because it is short-circuited. Also due to the short circuit, the $5\text{-}\Omega$ resistor, the voltage source, and the dependent current source are all in parallel. Hence, $i_x = 0$. At node a , $i_o = \frac{1\text{ V}}{5\text{ }\Omega} = 0.2\text{ A}$, and

$$R_N = \frac{v_o}{i_o} = \frac{1}{0.2} = 5\text{ }\Omega$$

To find I_N , we short-circuit terminals a and b and find the current i_{sc} , as indicated in Fig. 4.44(b). Note from this figure that the $4\text{-}\Omega$ resistor, the 10-V voltage source, the $5\text{-}\Omega$ resistor, and the dependent current source are all in parallel. Hence,

$$i_x = \frac{10}{4} = 2.5\text{ A}$$

At node a , KCL gives

$$i_{sc} = \frac{10}{5} + 2i_x = 2 + 2(2.5) = 7\text{ A}$$

Thus,

$$I_N = 7\text{ A}$$

1.2.7 Appendix. Derivation of Thevenin and Norton theorem

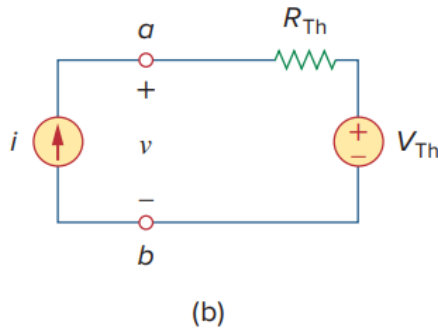
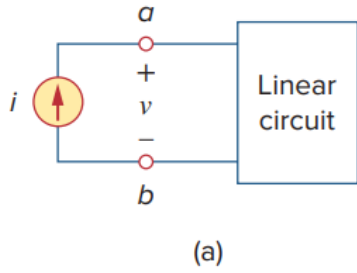


Figure 4.46

Derivation of Thevenin equivalent: (a) a current-driven circuit, (b) its Thevenin equivalent.

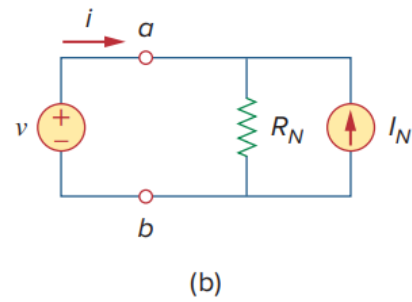
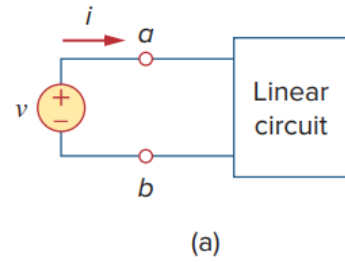


Figure 4.47

Derivation of Norton equivalent: (a) a voltage-driven circuit, (b) its Norton equivalent.

1. **Proof of Thevenin theorem.** In this section, we will prove Thevenin's and Norton's theorems using the superposition principle.

Consider the linear circuit in Fig. 4.46(a). It is assumed that the circuit contains resistors and dependent and independent sources. We have access to the circuit via terminals a and b , through which current from an external source is applied. Our objective is to ensure that the voltage–current relation at terminals a and b is identical to that of the Thevenin equivalent in Fig. 4.46(b). For the sake of simplicity, suppose the linear circuit in Fig. 4.46(a) contains two independent voltage sources v_{s1} and v_{s2} and two independent current sources i_{s1} and i_{s2} . We may obtain any circuit variable, such as the terminal voltage v , by applying superposition. That is, we consider the contribution due to each independent source including the external source i . By superposition, the terminal voltage v is

$$v = A_0 i + A_1 v_{s1} + A_2 v_{s2} + A_3 i_{s1} + A_4 i_{s2} \quad (4.13)$$

where A_0, A_1, A_2, A_3 , and A_4 are constants. Each term on the right-hand side of Eq. (4.13) is the contribution of the related independent source; that is, $A_0 i$ is the contribution to v due to the external current source i , $A_1 v_{s1}$ is the contribution due to the voltage source v_{s1} , and so on. We may collect terms for the internal independent sources together as B_0 , so that Eq. (4.13) becomes

$$v = A_0 i + B_0 \quad (4.14)$$

where $B_0 = A_1v_{s1} + A_2v_{s2} + A_3i_{s1} + A_4i_{s2}$. We now want to evaluate the values of constants A_0 and B_0 . When the terminals a and b are open-circuited, $i = 0$ and $v = B_0$. Thus, B_0 is the open-circuit voltage v_{oc} , which is the same as V_{Th} , so

$$B_0 = V_{Th} \quad (4.15)$$

When all the internal sources are turned off, $B_0 = 0$. The circuit can then be replaced by an equivalent resistance R_{eq} , which is the same as R_{Th} , and Eq. (4.14) becomes

$$v = A_0i = R_{Th}i \quad \Rightarrow \quad A_0 = R_{Th} \quad (4.16)$$

Substituting the values of A_0 and B_0 in Eq. (4.14) gives

$$v = R_{Th}i + V_{Th} \quad (4.17)$$

which expresses the voltage-current relation at terminals a and b of the circuit in Fig. 4.46(b). Thus, the two circuits in Fig. 4.46(a) and 4.46(b) are equivalent.

2. **Proof of Norton theorem.** When the same linear circuit is driven by a voltage source v as shown in Fig. 4.47(a), the current flowing into the circuit can be obtained by superposition as

$$i = C_0v + D_0 \quad (4.18)$$

where C_0v is the contribution to i due to the external voltage source v and D_0 contains the contributions to i due to all internal independent sources. When the terminals a - b are short-circuited, $v = 0$ so that

$$i = D_0 = -i_{sc},$$

where i_{sc} is the short-circuit current flowing out of terminal a , which is the same as the Norton current I_N , i.e.,

$$D_0 = -I_N \quad (4.19)$$

When all the internal independent sources are turned off, $D_0 = 0$ and the circuit can be replaced by an equivalent resistance R_{eq} (or an equivalent conductance $G_{eq} = 1/R_{eq}$), which is the same as R_{Th} or R_N . Thus, Eq. (4.19) becomes

$$i = \frac{v}{R_{Th}} - I_N \quad (4.20)$$

This expresses the voltage-current relation at terminals a - b of the circuit in Fig. 4.47(b), confirming that the two circuits in Fig. 4.47(a) and 4.47(b) are equivalent.

1.3 Practical Applications

1.3.1 Application 1. Source modeling

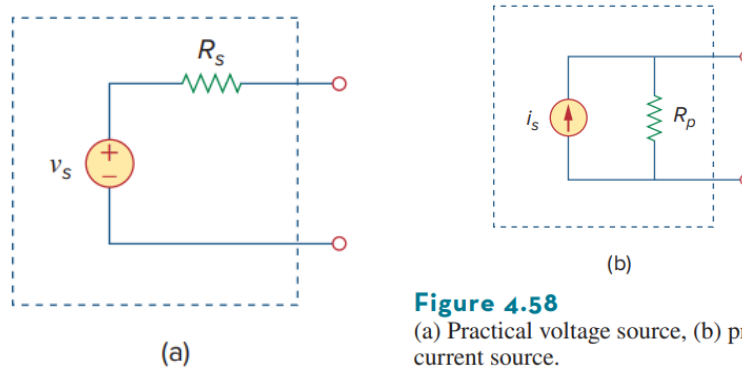


Figure 4.58

(a) Practical voltage source, (b) practical current source.

Source modeling provides an example of the usefulness of the Thevenin or the Norton equivalent. An active source such as a battery is often characterized by its Thevenin or Norton equivalent circuit. An ideal voltage source provides a constant voltage irrespective of the current drawn by the load, while an ideal current source supplies a constant current regardless of the load voltage. As Fig. 4.58 shows, practical voltage and current sources are not ideal, due to their *internal resistances* or *source resistances* R_s and R_p . They become ideal as $R_s \rightarrow 0$ and $R_p \rightarrow \infty$.

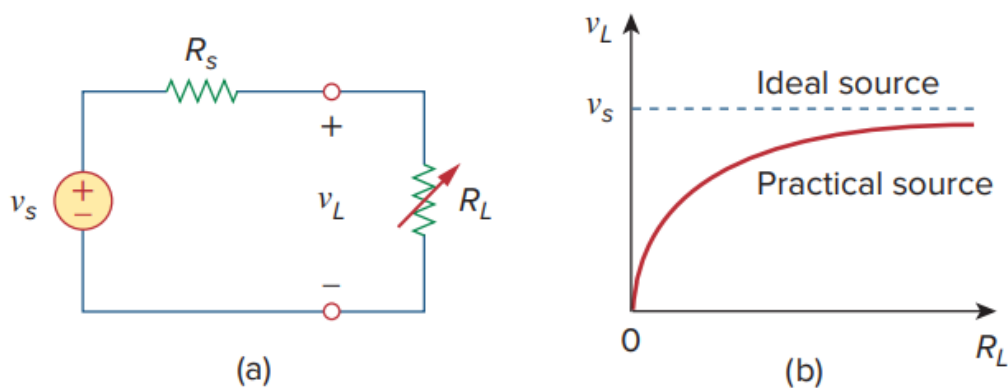


Figure 4.59

(a) Practical voltage source connected to a load R_L , (b) load voltage decreases as R_L decreases.

To show that this is the case, consider the effect of the load on voltage sources, as shown in Fig. 4.59(a). By the voltage division principle, the load voltage is

$$v_L = \frac{R_L}{R_s + R_L} v_s \quad (4.25)$$

As R_L increases, the load voltage approaches a source voltage v_s , as illustrated in Fig. 4.59(b). From Eq. (4.25), we should note that:

1. The load voltage will be constant if the internal resistance R_s of the source is zero or, at least, $R_s \ll R_L$. In other words, the smaller R_s is compared with R_L , the closer the voltage source is to being ideal.

2. When the load is disconnected (i.e., the source is open-circuited so that $R_L \rightarrow \infty$), $v_{oc} = v_s$. Thus, v_s may be regarded as the *unloaded source* voltage. The connection of the load causes the terminal voltage to drop in magnitude; this is known as the *loading effect*.

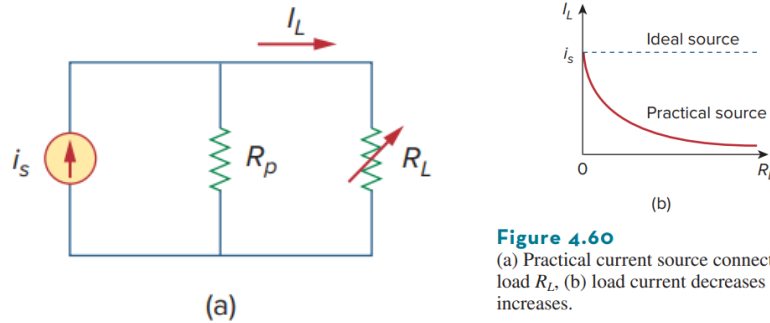


Figure 4.60
(a) Practical current source connected to a load R_L , (b) load current decreases as R_L increases.

The same argument can be made for a practical current source when connected to a load as shown in Fig. 4.60(a). By the current division principle,

$$i_L = \frac{R_p}{R_p + R_L} i_s \quad (4.26)$$

Figure 4.60(b) shows the variation in the load current as the load resistance increases. Again, we notice a drop in current due to the load (loading effect), and load current is constant (ideal current source) when the internal resistance is very large (i.e., $R_p \rightarrow \infty$ or, at least, $R_p \gg R_L$).

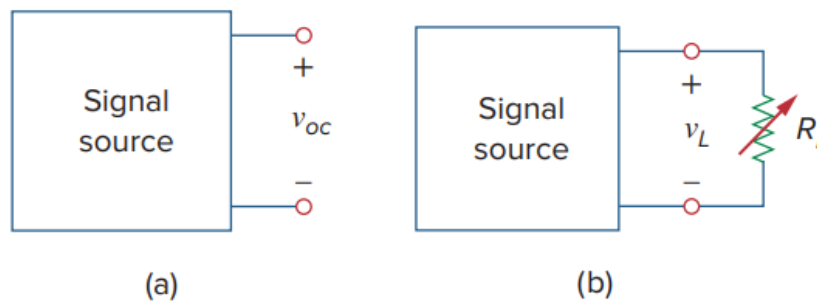


Figure 4.61
(a) Measuring v_{oc} , (b) measuring v_L .

Sometimes, we need to know the unloaded source voltage v_s and the internal resistance R_s of a voltage source. To find v_s and R_s , we follow the procedure illustrated in Fig. 4.61. First, we measure the open-circuit voltage v_{oc} as in Fig. 4.61(a) and set

$$v_s = v_{oc} \quad (4.27)$$

Then, we connect a variable load R_L across the terminals as in Fig. 4.61(b). We adjust the resistance R_L until we measure a load voltage of exactly one-half of the open-circuit voltage, $v_L = v_{oc}/2$, because now $R_L = R_{Th} = R_s$. At that point, we disconnect R_L and measure it. We set

$$R_s = R_L \quad (4.28)$$

For example, a car battery may have $v_s = 12\text{ V}$ and $R_s = 0.05\ \Omega$.

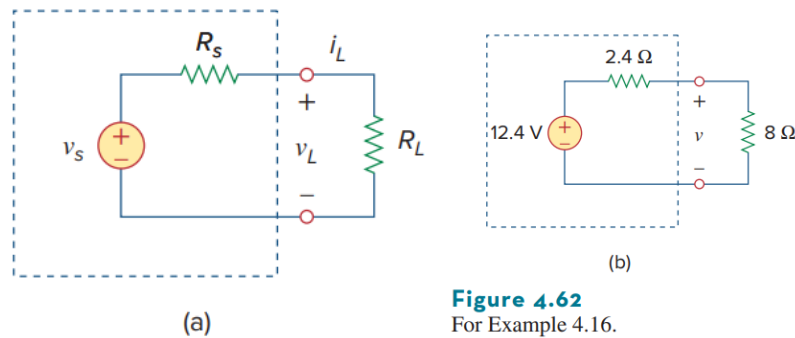
Example. Loading effect

Figure 4.62
For Example 4.16.

The terminal voltage of a voltage source is 12 V when connected to a 2-Ω load. When the load is disconnected, the terminal voltage rises to 12.4 V. (a) Calculate the source voltage v_s and internal resistance R_s . (b) Determine the voltage when an 8-Ω load is connected to the source.

Solution:

(a) We replace the source by its Thevenin equivalent. The terminal voltage when the load is disconnected is the open-circuit voltage,

$$v_s = v_{oc} = 12.4 \text{ V}$$

When the load is connected, as shown in Fig. 4.62(a), $v_L = 12 \text{ V}$ and $p_L = 2 \text{ W}$. Hence,

$$p_L = \frac{v_L^2}{R_L} \Rightarrow R_L = \frac{v_L^2}{p_L} = \frac{12^2}{2} = 72 \Omega$$

The load current is

$$i_L = \frac{v_L}{R_L} = \frac{12}{72} = \frac{1}{6} \text{ A}$$

The voltage across R_s is the difference between the source voltage v_s and the load voltage v_L , or

$$12.4 - 12 = 0.4 = R_s i_L, \quad R_s = \frac{0.4}{i_L} = 2.4 \Omega$$

(b) Now that we have the Thevenin equivalent of the source, we connect the 8-Ω load across the Thevenin equivalent as shown in Fig. 4.62(b). Using voltage division, we obtain

$$v = \frac{8}{8 + 2.4}(12.4) = 9.538 \text{ V}$$

1.3.2 Application 2. Resistance measurement

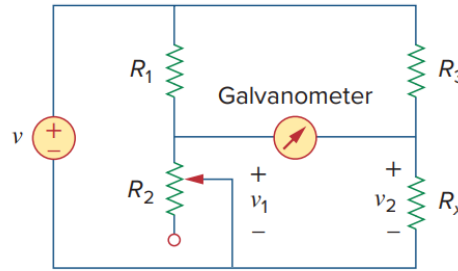


Figure 4.63

The Wheatstone bridge; R_x is the resistance to be measured.

Although the ohmmeter method provides the simplest way to measure resistance, more accurate measurement may be obtained using the Wheatstone bridge. While ohmmeters are designed to measure resistance in low, mid, or high range, a Wheatstone bridge is used to measure resistance in the mid range, say, between $1\ \Omega$ and $1\ \text{M}\Omega$. Very low values of resistances are measured with a *milliohmmeter*, while very high values are measured with a *Megger tester*.

The Wheatstone bridge (or resistance bridge) circuit is used in a number of applications. Here we will use it to measure an unknown resistance. The unknown resistance R_x is connected to the bridge as shown in Fig. 4.63. The variable resistance is adjusted until no current flows through the galvanometer, which is essentially a d'Arsonval movement operating as a sensitive current-indicating device like an ammeter in the microamp range. Under this condition $v_1 = v_2$, and the bridge is said to be *balanced*. Since no current flows through the galvanometer, R_1 and R_2 behave as though they were in series; so do R_3 and R_x . The fact that no current flows through the galvanometer also implies that $v_1 = v_2$. Applying the voltage division principle,

$$v_1 = \frac{R_2}{R_1 + R_2}v = v_2 = \frac{R_x}{R_3 + R_x}v \quad (4.29)$$

Hence, no current flows through the galvanometer when

$$\frac{R_2}{R_1 + R_2} = \frac{R_x}{R_3 + R_x} \Rightarrow R_2 R_3 = R_1 R_x$$

or

$$R_x = \frac{R_3}{R_1} R_2 \quad (4.30)$$

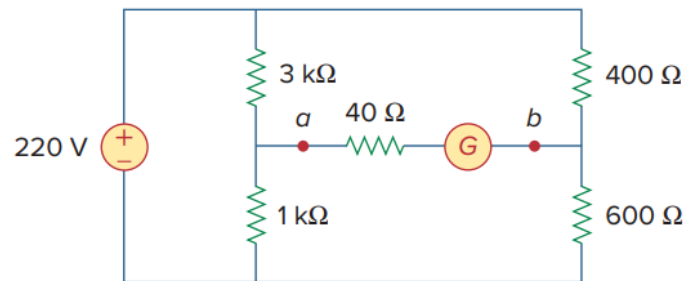
If $R_1 = R_3$, and R_2 is adjusted until no current flows through the galvanometer, then $R_x = R_2$.

How do we find the current through the galvanometer when the Wheatstone bridge is unbalanced? We find the Thevenin equivalent (V_{Th} and R_{Th}) with respect to the galvanometer terminals. If R_m is the resistance of the galvanometer, the current through it under the unbalanced condition is

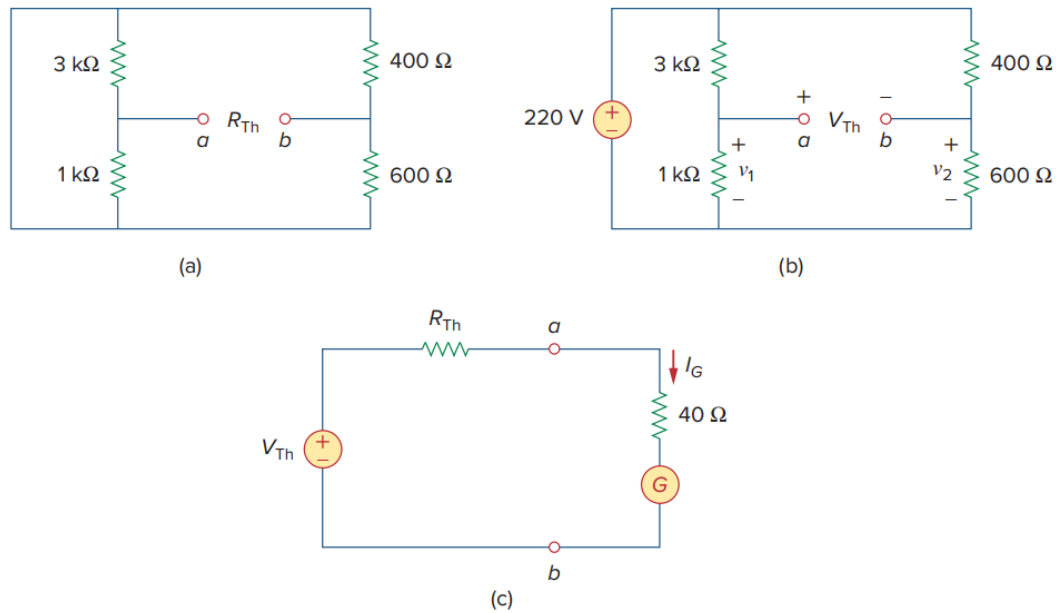
$$I = \frac{V_{\text{Th}}}{R_{\text{Th}} + R_m} \quad (4.31)$$

Example. Finding the current through galvanometer.

The circuit in Fig. 4.64 represents an unbalanced bridge. If the galvanometer has a resistance of $40\ \Omega$, find the current through the galvanometer.

**Figure 4.64**

Unbalanced bridge of Example 4.18.

**Figure 4.65**

For Example 4.18: (a) Finding R_{Th} , (b) finding V_{Th} , (c) determining the current through the galvanometer.

Solution:

We first need to replace the circuit by its Thevenin equivalent at terminals a and b . The Thevenin resistance is found using the circuit in Fig. 4.65(a). Notice that the $3\text{-k}\Omega$ and $1\text{-k}\Omega$ resistors are in parallel; so are the $400\text{-}\Omega$ and $600\text{-}\Omega$ resistors. The two parallel combinations form a series combination with respect to terminals a and b . Hence,

$$R_{Th} = 3000 \parallel 1000 + 400 \parallel 600 = \frac{3000 \times 1000}{3000 + 1000} + \frac{400 \times 600}{400 + 600} = 750 + 240 = 990\ \Omega$$

To find the Thevenin voltage, we consider the circuit in Fig. 4.65(b). Using the voltage division principle gives

$$v_1 = \frac{1000}{1000 + 3000}(220) = 55\text{ V}, \quad v_2 = \frac{600}{600 + 400}(220) = 132\text{ V}$$

Applying KVL around loop ab gives

$$-v_1 + V_{Th} + v_2 = 0 \quad \Rightarrow \quad V_{Th} = v_1 - v_2 = 55 - 132 = -77\text{ V}$$

Having determined the Thevenin equivalent, we find the current through the galvanometer using Fig. 4.65(c):

$$I_G = \frac{V_{Th}}{R_{Th} + R_m} = \frac{-77}{990 + 40} = -74.76\text{ mA}$$

The negative sign indicates that the current flows in the direction opposite to the one assumed, that is, from terminal b to terminal a .